Cressie–Read Power Divergence for Moment-Based Estimation: Hyperparameter and Finite-Sample Behavior

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Abstract

We study Cressie–Read power divergence (CRPD) estimation for moment-based models, focusing on finite-sample behavior. While generalized empirical likelihood estimators, dual to CRPD, are known to outperform generalized method of moments estimators in small to moderate samples, the power parameter is typically chosen arbitrarily by the researcher, serving mainly as an index. We interpret it as a hyperparameter that determines the loss function and governs the learning procedure, shaping the curvature of the objective and influencing finite-sample performance. Using second-order asymptotics, we show that it affects both the structural estimator and the associated Lagrange multipliers, governing robustness, bias, and sensitivity to sampling variation. Monte Carlo simulations illustrate how estimator performance varies with the choice of the power parameter and underlying distributional features, with implications for second-order bias and coverage distortion. An empirical illustration based on Owen (2001)'s classical example highlights the practical relevance of tuning the power parameter.

Keywords: Moment-based estimation; Cressie–Read power divergence; Generalized empirical likelihood; Finite-sample behavior; Hyperparameter.
JEL codes: C12, C13, C14.

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1 Introduction

First-order asymptotic analysis has been widely adopted for its simplicity and analytical tractability. Its effectiveness hinges on the availability of sufficient information relative to the complexity of the estimation problem. In many empirical settings, this requirement is not well met. In some cases, the available data are inherently limited by the nature of the problem.[1] Even when the nominal sample size is moderately large, the effective information per parameter may still be limited, particularly in settings where the number of moments or parameters is large relative to the sample size. As a result, estimation problems may behave more like small-sample regimes, where first-order approximations provide a poor guide to finite-sample performance (Huber, 1973).

These considerations motivate the use of Cressie–Read power divergence (CRPD) estimators (Cressie and Read, 1984) in moment-based estimation. CRPD estimators are closely connected to the generalized empirical likelihood (GEL) class through a well-known duality. The GEL literature has extensively documented favorable finite-sample properties relative to GMM (see, e.g., Owen (1988); Qin and Lawless (1994); Hansen et al. (1996); Kitamura and Stutzer (1997); Imbens et al. (1998); Newey and Smith (2004); Schennach (2007)), particularly in small to moderate samples. Through this duality, CRPD inherits these advantages, offering improved robustness by incorporating the finite-sample geometry of the moment conditions into the estimation procedure.

A key feature of the CRPD is the *power parameter,* which indexes a family of divergence functions. Different values of the power parameter label distinct members of the divergence family, with canonical choices corresponding to empirical likelihood, exponential tilting, or continuous updating. Typically, the power parameter is chosen *arbitrarily* by the researcher. Or, because the resulting estimators are first-order asymptotically equivalent under correct specification (see, e.g., Newey and Smith (2004)), the choice of the power parameter is generally regarded as asymptotically irrelevant.

In this paper, we show that the power parameter γ should not be selected arbitrarily, but instead can be interpreted as a *learning hyperparameter* in the sense of

[1] For example, when observations are costly, rare, or constrained by institutional or temporal factors—so that the sample size cannot be meaningfully increased

modern machine learning (ML) for two reasons. First, γ determines the loss function itself and therefore governs the *curvature of the learning objective*. This elevates γ beyond a conventional tuning constant to a parameter that shapes how the estimation problem is solved. We show that this choice is non-arbitrary: the CRPD criterion is well behaved as a function of γ and admits a unique minimizer, implying that the curvature of the objective is endogenously determined rather than imposed by convention.

Specifically, within the CRPD framework, γ controls the curvature of the penalty used to enforce moment conditions through empirical reweighting. Estimation begins with uniform weights assigned to all observations and then minimally perturbs these weights so that the sample moments satisfy the model's restrictions. The power parameter governs how such perturbations are penalized.[2]

Second, we show that γ does not enter the first-order asymptotic expansion, but plays a substantive role in the *second-order behavior* of both the structural estimator and the associated Lagrange multipliers. First-order asymptotics govern identification and consistency, whereas second-order terms determine finite-sample properties, including bias, robustness, and sensitivity to sampling variation (see, e.g., Rothenberg (1984); Newey and Smith (2004); Huber and Ronchetti (2009)). Because γ does not affect first-order identification, it cannot be interpreted as a structural parameter. Its role is instead to govern how learning proceeds in finite samples through its effect on higher-order behavior.

From this perspective, γ plays a role analogous to a hyperparameter: it does not alter the population target, but influences how the estimator responds to finite-sample features of the data. This interpretation connects the CRPD framework to ML ideas—such as tuning and model selection—while remaining within a standard moment-based econometric setting. In particular, γ can be viewed as a continuous parameter that governs the curvature of the objective and thereby the robustness and sensitivity of the estimator. Different values of γ lead to systematically different finite-sample behavior within a class of first-order equivalent estimators.

Bringing this perspective to the CRPD framework yields several practical advan-

[2]Negative values of γ induce aggressive down-weighting of observations associated with large moment residuals, enhancing robustness in the presence of heavy tails or outliers. Values of γ near zero yield logarithmic sensitivity, providing a smooth benchmark, while positive values generate gentler reweighting schemes that retain greater influence from all observations but may sacrifice robustness.

tages. Rather than fixing γ a priori, it can be selected in a data-driven manner, for example using cross-validation or out-of-sample criteria. This allows the estimator to adapt to the empirical environment and mitigates distortions arising from unreliable first-order approximations. In this sense, γ serves as a continuous control over finite-sample learning behavior, enabling the researcher to tailor robustness and sensitivity to the features of the data.

Our interpretation of γ as a learning hyperparameter admits a useful analogy to optimization procedures. In particular, it parallels the role of the step size (or learning rate) in stochastic gradient descent (SGD), where different choices can lead to the same asymptotic limit while producing different finite-sample behavior (see, e.g., Toulis and Airoldi (2017)). Although the step size does not affect the asymptotic target under standard conditions, it plays an important role in determining convergence speed, stability, and sensitivity to noise. Similarly, in the CRPD framework, the choice of γ leaves first-order identification and consistency unchanged, but governs second-order properties such as robustness, bias, and sensitivity to sampling variation. This analogy highlights a broader principle: parameters that are asymptotically irrelevant may nonetheless be important for finite-sample behavior.

More broadly, interpreting the CRPD power parameter as governing the curvature of the objective connects naturally to the literature on loss design in statistical estimation. Classical asymptotic theory organizes estimators into the M-, R-, and L-estimation frameworks (Serfling, 1980). While empirical likelihood and the broader CRPD/GEL family are not directly formulated within this taxonomy—being defined through constrained divergence minimization over probability weights—profiling out the weights and associated multipliers yields a representation analogous to an M-estimator. This connection allows their asymptotic behavior to be analyzed using standard M-estimation tools.

Within this perspective, our framework is most closely related to the robust M-estimation literature (Huber, 1964; Huber and Ronchetti, 2009), where robustness is achieved through the design of loss functions whose curvature governs the influence of extreme observations. In that setting, tuning constants regulate finite-sample robustness and efficiency without corresponding to structural parameters of the data-generating process. Our framework extends this idea to moment-based estimation: rather than modifying residual-level losses, the CRPD power parameter shapes the curvature of a divergence-based objective that enforces moment conditions through empirical reweighting. As in robust M-estimation, γ is not identified by population moments but plays a central role in determining finite-sample behavior.

Our analysis is closely related to Newey and Smith (2004), who develop higher-order asymptotic theory to compare the finite-sample performance of GMM and GEL estimators under fixed divergence choices. Their analysis treats the divergence as given and studies how different members of the GEL class affect higher-order behavior of the structural parameter. Accordingly, the power parameter enters only insofar as it indexes different estimators within the GEL family. While we adopt the same higher-order expansion framework, our focus is different: rather than comparing estimators conditional on a fixed divergence, we study the role of the divergence parameter itself. We show that γ affects finite-sample behavior without altering population identification. In particular, while first-order results do not depend on γ, it enters the second-order behavior of both the estimator and the associated Lagrange multipliers, thereby governing robustness and constraint enforcement in finite samples.

Our work is also related to Schennach (2007), who develops the exponentially tilted empirical likelihood (ETEL) framework within a divergence-based class indexed by a power parameter. While our setting differs, we share the perspective of treating the power parameter as a continuum-valued index linking different estimators. In contrast to Schennach (2007), who focuses on misspecified moment conditions, we maintain correct specification and show that the power parameter matters through second-order effects, shaping robustness and sensitivity even when estimators are first-order asymptotically equivalent.

The rest of the paper is organized as follows. In Section 2, we illustrate how the power parameter shapes the loss geometry. In Section 3, we develop the moment-based CRPD estimator and show how the structural parameter and the Lagrange multipliers depend on the power parameter at the second order, affecting finite-sample properties. In Section 4, we present Monte Carlo simulations. In Section 5, we discuss the implications of our proposed framework. In Section 6, we present an empirical application. In Section 7, we conclude.

2 The power parameter and the loss geometry

Let $\boldsymbol{\pi} = (\pi_1, \ldots, \pi_n)$ denote the model-implied probability vector, where $\pi_i > 0$ for all $i = 1, \ldots, n$ and $\sum_{i=1}^{n} \pi_i = 1$. Let $\mathbf{q} = (q_1, \ldots, q_n)$ denote a reference probability vector against which $\boldsymbol{\pi}$ is compared. For $\gamma \in \mathbb{R}\setminus\{-1, 0\}$, the CRPD criterion (Cressie and Read, 1984) is defined as

$$\mathscr{I}_\gamma(\boldsymbol{\pi}, \mathbf{q}) = \frac{1}{\gamma(\gamma+1)} \sum_{i=1}^{n} \pi_i \left[\left(\frac{\pi_i}{q_i} \right)^\gamma - 1 \right].$$

Observe that γ governs the curvature of the divergence penalty and therefore shapes how the estimator trades off fidelity to the reference measure against satisfaction of the moment conditions. For large positive values of γ, the divergence grows rapidly for small deviations, tightly constraining the optimized weights π_i around the reference distribution unless the sample moments strongly compel otherwise. As γ approaches zero, the divergence converges to the Kullback–Leibler form, allowing moderate deviations at relatively low cost. When γ is negative, the loss function becomes flatter around the center of the distribution while penalizing extreme distortions more sharply, effectively down-weighting observations that generate large moment residuals.

These changes in curvature have direct implications for estimation. In particular, the choice of γ determines how the estimator responds to empirical features such as heavy tails, skewness, or localized moment violations. Smaller or negative values of γ induce robustness by limiting the influence of extreme observations, while larger values enforce tighter adherence to the reference distribution. Consequently, different values of γ correspond to different second-order sensitivities of the estimator to local misspecification.

In particular, as the reference distribution, we adopt the *uniform distribution over the observed data points.* That is, we take $\mathbf{q} := \mathbf{i}/n$ because by the Glivenko–Cantelli theorem, the empirical distribution—which assigns equal mass to each realized sample point—converges uniformly to the true data-generating distribution. Hence, uniform weighting over the sample points (not over the population support) provides a consistent estimator of the true distribution.

The CRPD between $\boldsymbol{\pi}$ and $\mathbf{i}/n$ is defined as

$$\mathscr{I}_\gamma(\boldsymbol{\pi}, \mathbf{i}/n) = \frac{1}{\gamma(\gamma+1)} \sum_{i=1}^{n} \pi_i \left[(n\pi_i)^\gamma - 1 \right], \qquad \gamma \neq 0, -1, \tag{1}$$

with continuous extensions at $\gamma = 0$ and $\gamma = -1$. Up to an additive constant,

$\mathscr{I}_\gamma(\boldsymbol{\pi}, \mathbf{i}/n)$ admits the equivalent representations

$$\begin{cases} \mathscr{I}_\gamma(\boldsymbol{\pi}, \mathbf{i}/n)\big|_{\gamma=0} = \sum_{i=1}^{n} \pi_i \ln(\pi_i), \\ \mathscr{I}_\gamma(\boldsymbol{\pi}, \mathbf{i}/n)\big|_{\gamma=-1} = -\sum_{i=1}^{n} \ln(\pi_i), \end{cases}$$

where the case $\gamma = 0$ corresponds to the exponential tilting (ET) divergence, and the case $\gamma = -1$ corresponds to the empirical likelihood (EL) divergence.

Our objective is to formalize the interpretation of γ as a parameter governing the geometry of the learning objective, rather than as a fixed modeling choice imposed ex ante. Establishing existence and uniqueness of γ within the CRPD family is therefore essential to ensure that the learning objective is well defined and non-arbitrary.

A defining feature of the CRPD framework is that different values of γ correspond to distinct loss functions, each characterized by a different curvature and sensitivity to deviations between the empirical distribution and the reference distribution. In this sense, γ indexes a family of admissible learning objectives, with each value implying a different robustness profile, rather than acting as a tuning constant chosen independently of the estimation problem.

Observe that γ determines the form of the implied loss function. For instance, when the underlying measurements are approximately Gaussian, the squared-error (ℓ_2) loss is optimal in the sense of maximum likelihood. Within the CRPD family, this case corresponds to $\gamma = 1$, since

$$\mathscr{I}_\gamma(\pi, \mathbf{i}/n)\big|_{\gamma=1} = \frac{1}{2}\sum_{i=1}^{n} \pi_i \left[n\pi_i - 1\right] = \frac{1}{2}\left[n\sum_{i=1}^{n} \pi_i^2 - \underbrace{\sum_{i=1}^{n} \pi_i}_{=1}\right] = \frac{1}{2}\left[n\sum_{i=1}^{n}\left(\pi_i - \frac{1}{n}\right)^2\right],$$

which coincides with the squared Euclidean distance between the empirical weights and the uniform distribution. Other values of γ induce alternative loss geometries, emphasizing different aspects of deviation and robustness to misspecification as summarized in Table 1.

Interpreting the CRPD objective as a loss function clarifies why γ should be regarded as a legitimate parameter of the learning problem rather than a fixed modeling convention. While conventional applications treat γ as exogenously specified, such a practice implicitly fixes the curvature of the learning objective independently of the

Table 1: CRPD loss geometry and special cases

γ value	Core loss term	Geometry	Interpretation
$\gamma = -1$ (EL)	$-\sum_i \log(n\pi_i)$	Log barrier	Empirical likelihood. The objective diverges as $\pi_i \downarrow 0$, strongly discouraging vanishing weights and enforcing interior solutions.
$\gamma = 0$ (ET)	$\sum_i \pi_i \log(n\pi_i)$	Logarithmic	Exponential tilting. Corresponds to the Kullback–Leibler divergence from the uniform distribution; weights are smoothly reweighted and remain close to $1/n$ when the moment conditions are nearly satisfied.
$\gamma = 1$ (Pearson χ^2)	$\sum_i \pi_i^2$	Quadratic	Pearson χ^2 divergence. Equivalent to the squared Euclidean distance from the uniform distribution on the probability simplex; in moment space it corresponds to a quadratic (Hotelling-type) distance.

estimation problem. In contrast, our framework allows the curvature parameter γ to be treated as an object of analysis, whose role can be formally characterized.

3 Moment-based CRPD estimation

We consider CRPD estimation under moment conditions. Let $\{Z_i\}_{i=1}^n$ be an i.i.d. sample drawn from a probability distribution P on $(\mathcal{X}, \mathcal{A})$, where $Z_i = (Y_i, X_i)$ collects the observed outcome and covariates for unit i. Let $Z = (Y, X)$ denote a generic draw from P. Let $g : \mathcal{X} \times \Theta \to \mathbb{R}^q$ be a measurable moment function. We assume that the true parameter $\theta_0 \in \Theta$ satisfies the population moment condition $E\big[g(Z, \theta_0)\big] = 0$. Let $g_i(\theta) := g(Z_i, \theta) \in \mathbb{R}^q$ and define $\bar{g}(\theta) := \frac{1}{n}\sum_{i=1}^n g_i(\theta), \qquad \hat{\Omega}(\theta) := \frac{1}{n}\sum_{i=1}^n g_i(\theta) g_i(\theta)'$. Let $G_i(\theta) := \partial g_i(\theta)/\partial\theta'$, and define $\bar{G}(\theta) := \frac{1}{n}\sum_{i=1}^n G_i(\theta)$, $G_0 := E[G(Z, \theta_0)]$, $\Omega_0 := E[g(Z, \theta_0) g(Z, \theta_0)']$. Let $H_i(\theta)$ denote the collection of second derivatives of $g_i(\theta)$ w.r.t. θ (i.e. a $q \times p \times p$ tensor), and assume $E[\sup_{\theta\in\Theta} \|H(Z, \theta)\|] < \infty$.

3.1 Moment-based CRPD estimator

Fix $\gamma \in \Gamma$ with $\gamma \neq 0$ and $\gamma \neq -1$. Define the CRPD estimator:

$$(\hat{\theta}_\gamma, \hat{\boldsymbol{\pi}}_\gamma) = \arg\min_{\theta, \boldsymbol{\pi}} \mathscr{I}_\gamma(\boldsymbol{\pi}, \mathbf{i}/n) \quad \text{such that } \sum_{i=1}^n \pi_i = 1,\ \sum_{i=1}^n \pi_i g_i(\theta) = 0,\ \pi_i \geq 0.$$

The Lagrange function is

$$\begin{aligned} \mathcal{L}(\boldsymbol{\pi}, \theta, \lambda, \delta) &= \mathscr{I}_\gamma(\boldsymbol{\pi}, \mathbf{i}/n) + \lambda' \sum_{i=1}^n \pi_i g_i(\theta) + \delta\Big(\sum_{i=1}^n \pi_i - 1\Big) \\ &= \frac{1}{\gamma(\gamma+1)} \sum_{i=1}^n \pi_i \left[(n\pi_i)^\gamma - 1\right] + \lambda' \sum_{i=1}^n \pi_i g_i + \delta \left(\sum_{i=1}^n \pi_i - 1\right). \end{aligned} \tag{2}$$

The first derivative of (2) with respect to π_i is

$$\begin{aligned} \frac{\partial \mathcal{L}}{\partial \pi_i} &= \frac{(1+\gamma)(n\pi_i)^\gamma - 1}{\gamma(\gamma+1)} + \lambda' g_i(\theta) + \delta \\ &= \frac{1}{\gamma}(n\pi_i)^\gamma - \frac{1}{\gamma(\gamma+1)} + \lambda' g_i(\theta) + \delta. \end{aligned} \tag{3}$$

The first-order condition (FOC) is obtained by setting (3) equal to zero. Observe

$$0 = \frac{1}{\gamma}(n\pi_i)^\gamma - \frac{1}{\gamma(\gamma+1)} + \delta + \lambda' g_i(\theta).$$

Rearranging terms,

$$\frac{1}{\gamma}(n\pi_i)^\gamma = \frac{1}{\gamma(\gamma+1)} - \delta - \lambda' g_i(\theta).$$

Multiplying γ,

$$(n\pi_i)^\gamma = \frac{1}{\gamma+1} - \gamma\delta - \gamma\lambda' g_i(\theta).$$

Thus,

$$\pi_i = \frac{1}{n}\left(\frac{1}{\gamma+1} - \gamma\delta - \gamma\lambda' g_i(\theta)\right)^{1/\gamma}. \tag{4}$$

Before proceeding to the asymptotic analysis, it is useful to characterize the population solution of the CRPD problem under correct specification. The following lemma establishes the benchmark values of the parameters and multipliers around which the subsequent expansions are developed.

Lemma 3.1. (Population solution under correct specification) Under correct specification, the solution to the CRPD problem lies in a neighborhood of

$$\theta = \theta_0, \qquad \lambda = 0, \qquad \pi_i = 1/n, \qquad \delta_0 = -\frac{1}{\gamma+1}.$$

Proof. See Appendix A.1.

Lemma 3.1 identifies the population reference point for the CRPD optimization problem. Under correct specification, the population solution corresponds to uniform weights $\pi_i = 1/n$ with $\lambda = 0$ and $\delta_0 = -1/(\gamma+1)$. This characterization provides the baseline around which the sample estimators and Lagrange multipliers are expanded in the subsequent analysis.

3.2 Asymptotic analysis and finite sample properties

For each θ, the pair (λ', δ) is determined by the two constraints

$$\sum_{i=1}^{n} \pi_i(\theta, \lambda, \delta) = 1, \qquad \sum_{i=1}^{n} \pi_i(\theta, \lambda, \delta)\, g_i(\theta) = 0. \tag{5}$$

Define the stacked system

$$\Psi_n(\theta, \lambda, \delta) := \begin{pmatrix} \Psi_{n,1}(\theta, \lambda, \delta) \\ \Psi_{n,2}(\theta, \lambda, \delta) \end{pmatrix} = \begin{pmatrix} \frac{1}{n}\sum_{i=1}^{n} w_i(\theta, \lambda, \delta) - 1 \\ \frac{1}{n}\sum_{i=1}^{n} w_i(\theta, \lambda, \delta)\, g_i(\theta) \end{pmatrix},$$

where $w_i(\theta, \lambda, \delta) := \left(\frac{1}{\gamma+1} - \gamma\delta - \gamma\lambda' g_i(\theta)\right)^{1/\gamma}$, so that (5) is equivalent to $\Psi_n(\theta, \lambda, \delta) = 0$.

For each θ, let $(\hat\lambda(\theta), \hat\delta(\theta))$ denote a solution of $\Psi_n(\theta, \lambda, \delta) = 0$ (existence/uniqueness shown below) and define $\hat{\boldsymbol{\pi}}(\theta) = \boldsymbol{\pi}(\theta, \hat\lambda(\theta), \hat\delta(\theta))$. Define the profiled objective

$$L_n(\theta) := \mathscr{I}_\gamma\big(\hat{\boldsymbol{\pi}}(\theta), \mathbf{i}/n\big), \qquad \hat\theta \in \arg\min_{\theta\in\Theta} L_n(\theta).$$

We assume the followings.

Assumption 3.1. (Compactness) Θ is compact and $\theta_0 \in \mathrm{int}(\Theta)$.

Assumption 3.2. (Smoothness and envelope) $g(z, \theta)$ is continuously differentiable in θ for P-a.e. z, with Jacobian $G(z, \theta) := \partial g(z, \theta)/\partial\theta'$. Moreover,

$$E\Big[\sup_{\theta\in\Theta} \|g(Z, \theta)\|^2\Big] < \infty, \qquad E\Big[\sup_{\theta\in\Theta} \|G(Z, \theta)\|\Big] < \infty.$$

Assumption 3.3. (Identification) $E[g(Z, \theta_0)] = 0$ and $E[g(Z, \theta)] \neq 0$ for $\theta \neq \theta_0$. Let

$$G_0 := E[G(Z, \theta_0)], \qquad \Omega_0 := E[g(Z, \theta_0)g(Z, \theta_0)'],$$

and assume G_0 has full column rank and Ω_0 is positive definite.

Assumption 3.4. (Interior feasibility / positivity) There exists a neighborhood $\mathcal{N}$ of θ_0 and a constant $\kappa > 0$ such that, with probability approaching one, for each $\theta \in \mathcal{N}$ there exist (λ, δ) with $\|\lambda\| + \|\delta - \delta_0\| \leq \kappa$ satisfying

$$\frac{1}{\gamma + 1} - \gamma\delta - \gamma\lambda' g_i(\theta) \geq \kappa \quad \text{for all } i,$$

and the constraints (5).

Assumption 3.1 ensures that the parameter space is bounded and closed, which facilitates uniform convergence arguments and guarantees the existence of minimizers of the profiled criterion. The interior condition $\theta_0 \in \text{int}(\Theta)$ rules out boundary issues and permits local expansions around the true parameter. Assumption 3.2 provides the regularity required for both uniform laws of large numbers and Taylor expansions. Continuous differentiability of $g(z, \theta)$ in θ ensures that the multiplier system and the profiled objective are smooth functions of θ. The envelope conditions guarantee integrable bounds that allow uniform convergence of sample moments and control stochastic equicontinuity. Assumption 3.3 ensures that the population moment condition $E[g(Z, \theta)] = 0$ uniquely characterizes θ_0. The full column rank of G_0 guarantees local identification and invertibility of the Jacobian in the multiplier system, while the positive definiteness of Ω_0 ensures nondegeneracy of the moment covariance matrix. Together, these conditions imply both consistency and asymptotic normality of the estimator. Assumption 3.4 is a technical regularity condition ensuring that the implied weights are well-defined and that the multiplier system remains smooth in a neighborhood of θ_0. The uniform positivity bound prevents the implied probabilities from approaching the boundary of the simplex. This guarantees interior solutions, avoids singularities in the CRPD power transformation, and ensures that the mapping $(\theta, \lambda, \delta) \mapsto w_i(\theta, \lambda, \delta)$ is continuously differentiable on the relevant region.

Theorem 3.1. (Uniform convergence of the profiled objective) Fix the hyperparameter γ. Under Assumptions 1–4, there exists a deterministic function $L(\theta)$ such that

$$\sup_{\theta \in \Theta} \left| L_n(\theta) - L(\theta) \right| \xrightarrow{p} 0,$$

and $L(\theta)$ is uniquely minimized at θ_0.

Proof. See Appendix A.2.

Theorem 3.1 shows that the profiled sample objective $L_n(\theta)$ converges uniformly to a deterministic population criterion $L(\theta)$. This result ensures that the finite-sample objective approximates a well-defined population problem. Because $L(\theta)$ is uniquely minimized at θ_0, the minimizer of the sample objective will concentrate near the true parameter as the sample size increases. This property forms the basis for establishing consistency of the CRPD estimator in the subsequent results.

To analyze the profiled criterion $L_n(\theta)$, we must first ensure that the Lagrange multipliers solving the constraint system are well defined. In particular, we require that for θ in a neighborhood of the true value θ_0, the system of first-order conditions admits a unique solution for the multipliers. The following lemma establishes the local existence and uniqueness of $(\hat{\lambda}(\theta), \hat{\delta}(\theta))$. $(\hat{\lambda}(\theta), \hat{\delta}(\theta))$.

Lemma 3.2 (Local existence and uniqueness of multipliers)**.** Under Assumptions 3.1,3.2,3.3,3.4, there exists a neighborhood $\mathcal{N}$ of θ_0 such that, with probability approaching one, for every $\theta \in \mathcal{N}$ the system $\Psi_n(\theta, \lambda, \delta) = 0$ has a unique solution $(\hat{\lambda}(\theta), \hat{\delta}(\theta))$ satisfying $\|\hat{\lambda}(\theta)\| + \|\hat{\delta}(\theta) - \delta_0\| \leq \kappa$.

Proof. See Appendix A.3.

We can now characterize the large-sample behavior of the estimator obtained from minimizing $L_n(\theta)$. The following theorem establishes the consistency of $\hat{\theta}$.

Theorem 3.2 (Consistency of $\hat{\theta}$)**.** Under Assumptions 3.1,3.2,3.3, and Theorem 3.1, any measurable selection $\hat{\theta} \in \arg\min_{\theta \in \Theta} L_n(\theta)$ satisfies $\hat{\theta} \xrightarrow{p} \theta_0$.

Proof. See Appendix A.4.

The consistency of $\hat{\theta}$ further implies the consistency of the associated Lagrange multipliers. The following corollary formalizes this result.

Corollary 3.1 (Consistency of multipliers)**.** Suppose $\hat{\theta} \xrightarrow{p} \theta_0$. For each θ in a neighborhood of θ_0, let $(\lambda(\theta), \delta(\theta))$ denote the unique solution to $\Psi(\theta, \lambda, \delta) = 0$, and assume that the mapping $\theta \mapsto (\lambda(\theta), \delta(\theta))$ is continuous at θ_0. Define $(\hat{\lambda}, \hat{\delta}) := (\lambda(\hat{\theta}), \delta(\hat{\theta}))$. Then $(\hat{\lambda}, \hat{\delta}) \xrightarrow{p} (\lambda_0, \delta_0)$, where $(\lambda_0, \delta_0) = (\lambda(\theta_0), \delta(\theta_0))$.

Proof. See Appendix A.5

Having established the consistency of the estimator and the associated multipliers, we next characterize their asymptotic behavior. The following result provides the first-order expansion of $\hat{\lambda}$ and the convergence rate of $\hat{\delta}$.

Theorem 3.3 (First-order expansion for $\hat{\lambda}(\hat{\theta})$ and rate for $\hat{\delta}(\hat{\theta})$)**.** Under Assumptions 3.1,3.2,3.3,3.4 and Theorem 3.2, let $\hat{\lambda} := \hat{\lambda}(\hat{\theta})$ and $\hat{\delta} := \hat{\delta}(\hat{\theta})$. Then, under correct specification,

$$\hat{\lambda} = \hat{\Omega}(\theta_0)^{-1}\bar{g}(\theta_0) + o_p(n^{-1/2}), \qquad \hat{\delta} - \delta_0 = O_p(n^{-1}),$$

and

$$\sqrt{n}\,\hat{\lambda} \ \xrightarrow{d}\ N(0, \Omega_0^{-1}).$$

Proof. See Appendix A.6.

While Theorem 3.3 establishes the convergence rate of $\hat{\delta}$, its limiting distribution requires a finer expansion. The following theorem characterizes the nondegenerate limit of the probability multiplier.

Theorem 3.4 (Nondegenerate limit for the probability multiplier)**.** Assume the conditions of Theorem 3.3. Suppose additionally that a third-order remainder control holds so that the quadratic approximation in $t_i(\theta)$ is valid at the n^{-1} scale. Then

$$n(\hat{\delta} - \delta_0) = -\frac{\gamma+1}{2}\left(\sqrt{n}\,\bar{g}(\theta_0)\right)'\Omega_0^{-1}\left(\sqrt{n}\,\bar{g}(\theta_0)\right) + o_p(1),$$

and hence

$$n(\hat{\delta} - \delta_0) \ \xrightarrow{d}\ -\frac{\gamma+1}{2}\,\chi_q^2,$$

where χ_q^2 denotes a chi-square random variable with q degrees of freedom.

Proof. See Appendix A.7.

With the expansions for the multipliers established, we now derive the asymptotic distribution of the estimator.

Theorem 3.5 (Asymptotic normality of $\hat{\theta}$)**.** Under Assumptions 3.1,3.2,3.3,3.4 and Theorem 3.2. Then

$$\sqrt{n}\,(\hat{\theta}-\theta_0) = -(G_0'\Omega_0^{-1}G_0)^{-1}G_0'\Omega_0^{-1}\sqrt{n}\,\bar{g}(\theta_0) + o_p(1),$$

and hence

$$\sqrt{n}\,(\hat{\theta}-\theta_0) \xrightarrow{d} N\big(0,\ (G_0'\Omega_0^{-1}G_0)^{-1}\big)\,.$$

Proof. See Appendix A.8.

To facilitate the second-order analysis, we first derive a second-order expansion of the moment equation implied by the CRPD constraints.

Lemma 3.3 (Second-order expansion of the moment equation)**.** Let $(\hat{\theta}, \hat{\lambda}, \hat{\delta})$ satisfy the CRPD constraints:

$$\frac{1}{n}\sum_{i=1}^{n} w_i(\hat{\theta}, \hat{\lambda}, \hat{\delta}) = 1, \qquad \frac{1}{n}\sum_{i=1}^{n} w_i(\hat{\theta}, \hat{\lambda}, \hat{\delta})\, g_i(\hat{\theta}) = 0.$$

Then, uniformly on events of probability approaching one,

$$0 = \bar{g}(\hat{\theta}) - \hat{\Omega}(\hat{\theta})\hat{\lambda} + \frac{1-\gamma}{2}\cdot\frac{1}{n}\sum_{i=1}^{n}\big(t_i(\hat{\theta}, \hat{\lambda}, \hat{\delta})\big)^2 g_i(\hat{\theta}) + o_p(n^{-1}). \tag{6}$$

Proof. See Appendix A.9.

Using the second-order expansion of the moment equation in Lemma 3.3, we can now refine the first-order result for the multiplier. The following theorem provides the second-order expansion of $\hat{\lambda}$.

Theorem 3.6 (Second-order expansion for $\hat{\lambda}$)**.** Under the conditions of Lemma 3.3 and the first-order expansion

$$\hat{\lambda} = \hat{\Omega}(\theta_0)^{-1}\bar{g}(\theta_0) + o_p(n^{-1/2}),$$

we have

$$\hat{\lambda} = \hat{\Omega}(\theta_0)^{-1}\bar{g}(\theta_0) \;+\; \frac{1}{n}\,B_{\lambda,n}(\gamma) \;+\; o_p(n^{-1}), \tag{7}$$

where the sample-dependent second-order correction is

$$B_{\lambda,n}(\gamma) := \frac{1-\gamma}{2}\,\hat{\Omega}(\theta_0)^{-1}\left[\frac{1}{n}\sum_{i=1}^{n}\Big((\hat{\Omega}(\theta_0)^{-1}\sqrt{n}\,\bar{g}(\theta_0))'g_i(\theta_0)\Big)^2 g_i(\theta_0)\right], \tag{8}$$

and $B_{\lambda,n}(\gamma) = O_p(1)$.

Proof. See Appendix A.10.

Finally, combining the first-order asymptotic representation of $\hat{\theta}$ with the second-order expansion of the multiplier $\hat{\lambda}$, we can now derive the second-order expansion of the CRPD estimator.

Theorem 3.7. (Second-order expansion for $\hat{\theta}$) Assume the conditions of Theorems 3.5 and 3.6. Then the CRPD estimator $\hat{\theta}$ admits the expansion

$$\hat{\theta} - \theta_0 = \big(G_0'\Omega_0^{-1}G_0\big)^{-1}G_0'\Omega_0^{-1}\bar{g}(\hat{\theta}) + \frac{1}{n}B_{\theta,n}(\gamma) + o_p(n^{-1}),$$

where $\bar{g}(\theta) = \frac{1}{n}\sum_{i=1}^{n} g_i(\theta)$ and $B_{\theta,n}(\gamma) = \big(G_0'\Omega_0^{-1}G_0\big)^{-1}\Delta_n(\gamma) = O_p(1)$, with $\Delta_n(\gamma) = O_p(1)$ collects all order-n^{-1} terms arising from the second-order expansion of the Lagrange multiplier, the curvature term involving $\bar{H}(\theta_0)$, and plug-in effects from replacing sample matrices with their population limits.

Proof. See Appendix A.11.

Our asymptotic analysis characterizes the large-sample behavior of the CRPD estimator and its associated multipliers. The estimator $\hat{\theta}$ is consistent and asymptotically normal, sharing the same first-order limit as standard moment-based estimators. The power parameter γ does not affect the leading asymptotic distribution, but it appears in the order-n^{-1} bias term arising from the second-order expansions of the multipliers and the estimator. Consequently, the choice of γ influences finite-sample performance while leaving first-order asymptotic efficiency unchanged.

4 Monte Carlo simulations

We conduct Monte Carlo simulations with 1,000 replications. Each replication uses sample sizes $n \in \{25, 50, 100\}$ to represent small and moderate sample settings commonly encountered in empirical applications. The objective is to evaluate finite-sample behavior across different tail environments and to examine how the CRPD power parameter γ affects higher-order performance while preserving first-order asymptotic properties.

We consider estimation of the parameter vector $\theta = (\mu, \sigma^2)$ using moment conditions based on the first two central moments. In addition, we impose a third moment condition on skewness. Specifically, the moment conditions are

$$\mathrm{E}[X - \mu] = 0, \qquad \mathrm{E}\big[(X - \mu)^2 - \sigma^2\big] = 0, \qquad \mathrm{E}\big[(X - \mu)^3\big] = 0.$$

The third moment restriction is a *valid* population moment under the data-generating processes considered below. Because the Student-t and normal distributions are symmetric, the population third central moment equals zero. Thus, the third restriction does not introduce an additional structural parameter, but it renders the system overidentified. This avoids the degeneracy that can arise when too few moment conditions are imposed, under which the CRPD objective may admit a trivial solution with uniform weights.

Data are generated from symmetric t and normal distributions. Let $Z_i \sim t_\nu$ with mean zero and degrees of freedom $\nu \in \{5, 15\}$. The variance of the t distribution is $\nu/(\nu - 2)$, while the variance of the normal distribution is set to 1. Smaller values of ν correspond to heavier tails, whereas larger values approximate the normal distribution. This allows us to evaluate estimator performance across a range of tail behaviors.

We evaluate the estimator over a grid $\gamma \in [-1, 1]$ with step size 0.25 to illustrate how finite-sample performance varies with the choice of γ. For each replication and each candidate value of γ, the CRPD estimator is computed by searching over grids for (μ, σ^2) that cover a data-driven region around the sample mean and variance. For every candidate triple (μ, σ^2, γ), we solve for the associated multipliers (λ, δ) and the implied optimal weights $\{\pi_i\}$, and evaluate the CRPD objective. Each replication yields the estimators $\hat{\mu}(\gamma)$ and $\hat{\sigma}^2(\gamma)$, together with the associated Lagrange multipliers and implied weights.

We evaluate simulation performance using both higher-order and first-order measures. Finite-sample performance is evaluated using bias, mean squared error (MSE), and coverage distortion. Bias is defined as the average of $\hat{\mu} - \mu_0$ across replications.

MSE denotes the mean squared error of $\hat{\mu}$.[3] Coverage distortion is defined as the deviation of the empirical coverage probability from the nominal 95% level.[4] Bias, MSE, and coverage distortion are interpreted as higher-order performance measures because they reflect finite-sample deviations from the first-order asymptotic approximation.

First-order behavior is assessed by comparing the empirical standard deviation of $\hat{\mu}$ with its average estimated standard error. The empirical SD is the standard deviation of $\hat{\mu}$ across replications, and the mean SE is the average of the estimated standard errors. The SD–SE ratio denotes the ratio of the empirical SD to the mean SE. The empirical SD, mean SE, and their ratio summarize first-order behavior by capturing the variability of the estimator and the accuracy of the first-order standard error approximation.

The simulation results for small samples are reported in Tables 2, 3, and 4 for $n \in \{25, 50\}$ and Tables S.1, S.2 in Supplement, and the results for a larger sample size of $n = 100$ are reported in Tables S.3 and S.4 in Supplement. Table 2 summarizes both first- and second-order performance metrics. We observe that the higher-order measures (bias, MSE, and coverage distortion) and the first-order diagnostic SD–SE ratio vary with the value of γ. This indicates that when the sample size is small, both first- and higher-order measures fluctuate. The first-order measures vary because the sample size has not yet reached the asymptotic regime, while the higher-order measures vary because finite-sample effects remain non-negligible when n is small. These findings suggest that finite-sample considerations are important and that estimator performance depends on the choice of γ.

The optimal value of γ depends on the tail behavior of the data-generating process (DGP). Under heavier tails (e.g., t_5), negative values of γ tend to produce smaller MSE. As the tails become lighter and approach the normal distribution, positive values of γ tend to yield smaller MSE and smaller coverage distortions. With an appropriate choice of γ, coverage improves as n moves toward the asymptotic regime. In particular, coverage distortion decreases as the sample size increases from $n = 25$

[3] Although MSE decomposes as $\mathrm{Var}(\hat{\theta}) + \mathrm{Bias}(\hat{\theta})^2$, the leading variance term is governed by the first-order asymptotic approximation and is invariant to γ in our setting. Differences in MSE across values of γ therefore arise primarily through bias, which is driven by higher-order terms. For this reason, MSE is interpreted here as a finite-sample (higher-order) performance measure.

[4] Coverage distortion measures the deviation of the empirical coverage probability from the nominal confidence level. Because first-order asymptotic theory guarantees correct coverage only in the limit, deviations from the nominal level arise from higher-order terms in finite samples. Coverage distortion therefore reflects higher-order accuracy of the estimator and the associated standard error.

to $n = 50$. For example, under the t_5 distribution, $\gamma = -1$ yields the lowest MSE when $n = 25$, with coverage distortion -0.1090 and an SD–SE ratio of 1.3841. When $n = 50$, the lowest MSE occurs at $\gamma = -0.75$, with coverage distortion -0.0373 and an SD–SE ratio of 1.1584, both of which are smaller than their counterparts at $n = 25$.

Figure 1 plots second-order bias and coverage distortion for DGPs given by a t_5 distribution and a normal distribution with $n = 50$.[5] The filled circle indicates the value of γ that minimizes the absolute value of the metric. For the t_5 DGP, this optimal γ tends to be negative for both bias and coverage distortion, reflecting the need to downweight outliers under heavy tails. In contrast, for the normal DGP, the optimal γ is positive across both metrics. This pattern suggests that γ acts as a data-dependent tuning parameter, adjusting to distributional features to improve finite-sample performance.

Tables 3, S.2, and 4 report the estimates of μ, the associated Lagrange multipliers, and summary statistics of the estimated weights $\boldsymbol{\pi}$. Because the model is overidentified through the additional skewness restriction, the Lagrange multipliers associated with the moment conditions are generally nonzero. Consequently, the estimated weights $\boldsymbol{\pi}$ do not collapse to the uniform distribution. Depending on the characteristics of the data and the value of γ, the multipliers λ and δ can occasionally take relatively large values, reflecting the degree of adjustment in the probability weights required to satisfy the moment restrictions under the chosen divergence parameter γ.

These findings are consistent with the theoretical results developed earlier. In particular, the first-order asymptotic distribution of the CRPD estimator does not depend on the value of γ, while higher-order terms governing finite-sample bias and coverage distortion may vary with γ. The simulation evidence reflects this structure: measures related to first-order behavior, such as the empirical SD and mean SE, remain relatively stable across different values of γ, whereas higher-order measures such as bias, MSE, and coverage distortion display noticeable variation. This pattern confirms that the choice of γ primarily influences higher-order finite-sample properties rather than the leading asymptotic distribution.

The dependence of the optimal γ on tail behavior can be interpreted through

[5] We use $n = 50$ in this figure to ensure that first-order asymptotic approximations are reasonably accurate. When $n = 25$, the sample size appears too small for first-order behavior to be well approximated. At $n = 50$, first-order effects are more stable, while second-order effects remain non-negligible.

the way the divergence parameter shapes the implied probability weights. Negative values of γ tend to assign relatively more weight to observations with larger moment residuals, which can help stabilize estimation in heavy-tailed environments. In contrast, positive values of γ penalize large residuals more strongly, leading to more concentrated weights that perform better when the underlying distribution is closer to normal. As a result, the value of γ effectively controls how aggressively the estimator adjusts the implied probabilities in response to deviations from the moment conditions.

Overall, the simulation results highlight the importance of the divergence parameter in finite samples. Although all values of γ share the same first-order asymptotic properties, finite-sample performance can differ substantially depending on the choice of γ and the tail behavior of the underlying distribution. This suggests that selecting γ with attention to finite-sample considerations can lead to meaningful improvements in estimation accuracy and coverage performance.

A caveat is that the relationship between MSE and γ is generally non-monotonic. Consequently, the value of γ that minimizes MSE may occur at an interior point rather than at the boundary of the admissible range. This highlights the importance of evaluating the estimator over a sufficiently wide window of γ values; restricting the search to a narrow range could fail to identify the value of γ that delivers the best finite-sample performance.

5 Discussion

In classical regularization problems, a hyperparameter is chosen to balance bias and variance, typically by minimizing MSE. For an estimator indexed by a tuning parameter ζ, one often has

$$\mathrm{MSE}(\zeta) = \mathrm{Var}(\hat{\theta}_\zeta) + \mathrm{Bias}(\hat{\theta}_\zeta)^2.$$

The optimal value of ζ therefore reflects a bias–variance tradeoff. In the CRPD framework, however, the role of the power parameter γ is structurally different.

The first-order asymptotic distribution of $\hat{\theta}$ does not depend on γ:

$$\sqrt{n}(\hat{\theta} - \theta_0) \ \xrightarrow{d} \ N(0, V),$$

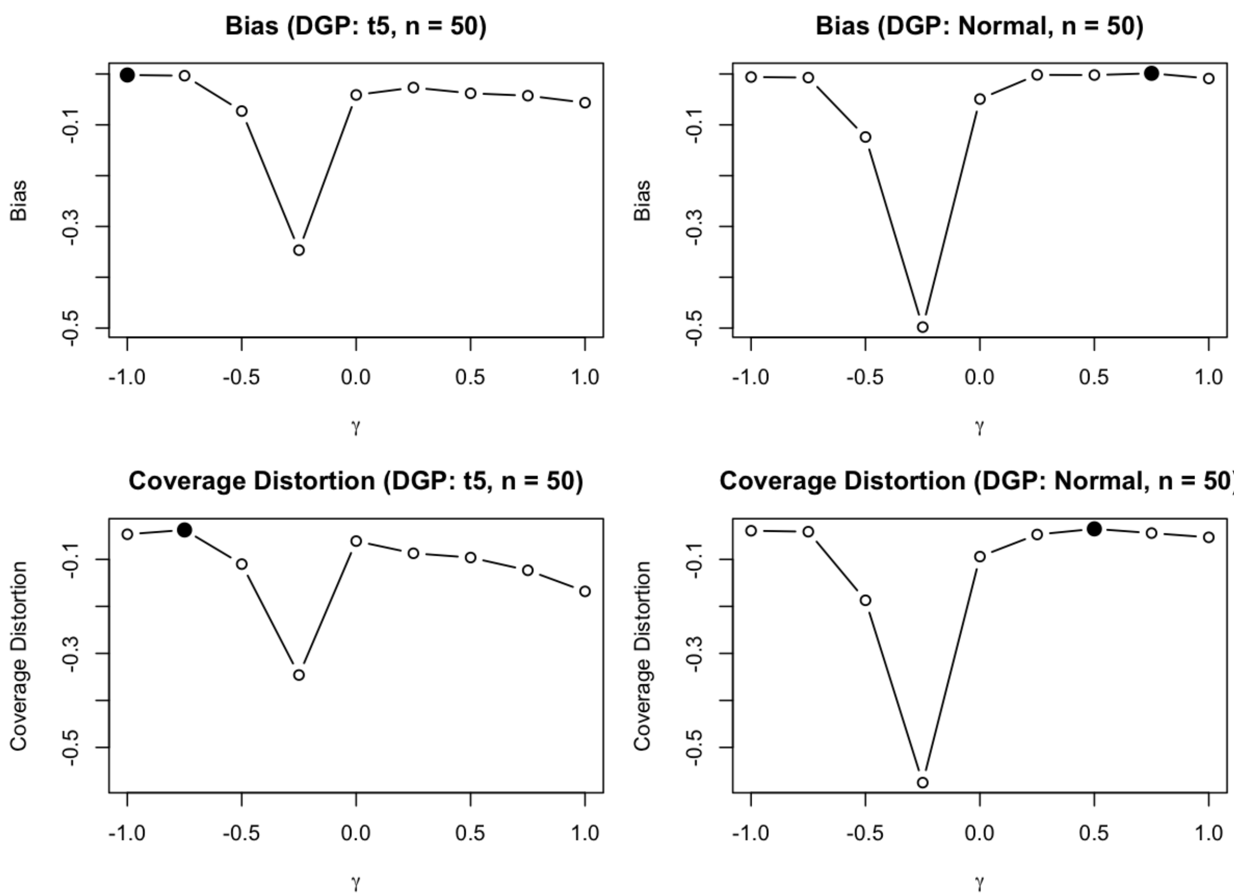
so the leading variance component in the MSE, V/n, is invariant to γ.

Table 2: Simulation performance metrics: second-order and first-order measures

DGP	n	γ	Second-order measure			First-order measure		
			Bias	MSE	Coverage distortion	Empirical SD	Mean SE	Ratio
t_5	25	-1.00	-0.0174	0.0923	-0.1090	0.3034	0.2192	1.3841
		-0.75	-0.0354	0.1089	-0.1374	0.3283	0.2182	1.5044
		-0.50	-0.2125	0.2776	-0.2806	0.4823	0.2309	2.0893
		-0.25	-0.4166	0.4440	-0.4075	0.5203	0.2473	2.1039
		0.00	-0.1347	0.2138	-0.1860	0.4425	0.2317	1.9100
		0.25	-0.0007	0.1109	-0.0821	0.3331	0.2221	1.4999
		0.50	-0.0205	0.1412	-0.1155	0.3754	0.2215	1.6947
		0.75	-0.0436	0.1561	-0.1388	0.3929	0.2241	1.7533
		1.00	-0.0685	0.1822	-0.1764	0.4215	0.2235	1.8859
	50	-1.00	-0.0020	0.0363	-0.0465	0.1906	0.1638	1.1641
		-0.75	-0.0034	0.0360	-0.0373	0.1898	0.1638	1.1584
		-0.50	-0.0730	0.1130	-0.1100	0.3283	0.1677	1.9574
		-0.25	-0.3466	0.3863	-0.3460	0.5162	0.1826	2.8265
		0.00	-0.0412	0.0712	-0.0610	0.2638	0.1664	1.5855
		0.25	-0.0267	0.0853	-0.0870	0.2910	0.1669	1.7434
		0.50	-0.0380	0.1078	-0.0960	0.3263	0.1680	1.9421
		0.75	-0.0427	0.1155	-0.1232	0.3374	0.1680	2.0085
		1.00	-0.0565	0.1334	-0.1680	0.3611	0.1697	2.1279
Normal	25	-1.00	-0.0180	0.0540	-0.1210	0.2317	0.1703	1.3608
		-0.75	-0.0416	0.0693	-0.1350	0.2602	0.1715	1.5172
		-0.50	-0.2743	0.2592	-0.4060	0.4291	0.1884	2.2778
		-0.25	-0.4780	0.4040	-0.5550	0.4192	0.2079	2.0162
		0.00	-0.1594	0.1771	-0.2230	0.3896	0.1864	2.0904
		0.25	-0.0003	0.0506	-0.1040	0.2250	0.1732	1.2992
		0.50	0.0032	0.0536	-0.1110	0.2317	0.1720	1.3465
		0.75	0.0036	0.0529	-0.1130	0.2300	0.1711	1.3439
		1.00	0.0018	0.0587	-0.1140	0.2423	0.1714	1.4141
	50	-1.00	-0.0060	0.0219	-0.0390	0.1478	0.1311	1.1275
		-0.75	-0.0071	0.0257	-0.0410	0.1602	0.1314	1.2189
		-0.50	-0.1240	0.1345	-0.1870	0.3453	0.1369	2.5218
		-0.25	-0.4980	0.4608	-0.5750	0.4615	0.1575	2.9302
		0.00	-0.0493	0.0653	-0.0940	0.2509	0.1345	1.8646
		0.25	-0.0017	0.0222	-0.0470	0.1491	0.1313	1.1355
		0.50	-0.0021	0.0219	-0.0350	0.1480	0.1321	1.1203
		0.75	0.0013	0.0244	-0.0440	0.1563	0.1322	1.1820
		1.00	-0.0089	0.0277	-0.0530	0.1661	0.1328	1.2514

Note: Results are based on 1,000 Monte Carlo replications. DGP denotes the data-generating process. n denotes the number of observations and γ the power parameter. Bias is the average of $\hat{\theta} - \theta_0$ across replications. MSE denotes the mean squared error of $\hat{\theta}$. Coverage distortion is the deviation of the empirical coverage probability from the nominal level 95%. Empirical SD is the standard deviation of $\hat{\theta}$ across replications. Mean SE is the average of the estimated standard errors. Ratio denotes the ratio of the empirical SD to the mean SE. Bias, MSE, and coverage distortion are reported as second-order performance measures because they reflect finite-sample deviations from the first-order asymptotic approximation. Empirical SD, mean SE, and their ratio summarize first-order behavior, as they evaluate the variability of the estimator and the accuracy of the first-order standard error approximation.

Figure 1: Second-order bias and coverage distortion ($n = 50$)



Note: Results are based on 1,000 Monte Carlo replications.DGP denotes the data-generating process, which is either a t distribution with 15 degrees of freedom or a normal distribution in this figure. n denotes the number of observations and γ the power parameter. Bias is the average of $\hat{\theta} - \theta_0$ across replications. Coverage distortion is the deviation of the empirical coverage probability from the nominal level 95%. The filled circle indicates the value of γ that minimizes the absolute value of the metric.

Table 3: Simulation results (t_5)

n	γ	$\hat{\theta}$	Lagrange multipliers λ_1	λ_2	λ_3	δ	$\boldsymbol{\pi}$ Minimum	Q1	Median	Mean	Q3	Maximum
25	-1.00	-0.0174	-0.0648	-0.0797	0.1092	-1.00E+04	0.0208	0.0337	0.0374	0.0400	0.0449	0.0642
		(0.3034)	(1.2644)	(1.3803)	(2.0990)	(1.42E-05)	(0.0122)	(0.0051)	(0.0043)	(3.88E-09)	(0.0042)	(0.0336)
	-0.75	-0.0354	-2.92E+10	-2.01E+10	1.09E+10	-1.70E+09	0.0192	0.0331	0.0367	0.0388	0.0437	0.0600
		(0.3283)	(5.15E+11)	(3.44E+11)	(1.85E+11)	(3.46E+10)	(0.0128)	(0.0075)	(0.0076)	(0.0067)	(0.0087)	(0.0344)
	-0.50	-0.2125	-3.38E+12	-1.91E+12	8.70E+11	-2.47E+11	0.0124	0.0258	0.0288	0.0304	0.0345	0.0489
		(0.4823)	(9.04E+13)	(4.52E+13)	(1.78E+13)	(6.65E+12)	(0.0129)	(0.0152)	(0.0167)	(0.0171)	(0.0198)	(0.0459)
	-0.25	-0.4166	-1.66E+12	-1.87E+12	1.14E+12	-2.04E+11	0.0088	0.0178	0.0197	0.0211	0.0231	0.0398
		(0.5203)	(1.53E+13)	(1.82E+13)	(1.09E+13)	(2.01E+12)	(0.0126)	(0.0177)	(0.0194)	(0.0200)	(0.0227)	(0.0723)
	0.00	-0.1347	4.51E+01	-8.70E+01	2.70E+01	-8.82E+01	0.0155	0.0296	0.0326	3.30E+74	0.0370	8.24E+75
		(0.4425)	(3.40E+02)	(5.26E+02)	(1.51E+02)	-6.68E+02	(0.0138)	(0.0140)	(0.0151)	-1.04E+76	(0.0172)	(2.61E+77)
	0.25	-0.0007	0.0070	-0.0190	-0.0024	-0.8618	0.0164	0.0352	0.0388	0.0400	0.0446	0.0598
		(0.3331)	(0.4426)	(0.0952)	(0.1307)	(0.1263)	(0.0137)	(0.0053)	(0.0041)	(5.43E-10)	(0.0038)	(0.0354)
	0.50	-0.0205	0.0271	-0.0188	-0.0036	-0.7522	0.0160	0.0347	0.0385	0.0400	0.0447	0.0612
		(0.3754)	(0.4858)	(0.0874)	(0.1254)	(0.1810)	(0.0139)	(0.0064)	(0.0045)	(2.24E-10)	(0.0038)	(0.0367)
	0.75	-0.0436	0.0498	-0.0173	-0.0040	-0.6803	0.0157	0.0341	0.0385	0.0400	0.0452	0.0610
		(0.3929)	(0.5223)	(0.0872)	(0.1217)	(0.2200)	(0.0141)	(0.0075)	(0.0044)	(1.26E-10)	(0.0042)	(0.0326)
	1.00	-0.0685	0.0919	-0.0200	-0.0051	-0.6494	0.0157	0.0331	0.0383	0.0400	0.0460	0.0627
		(0.4215)	(0.6059)	(0.0800)	(0.1294)	(0.2983)	(0.0142)	(0.0084)	(0.0045)	(1.44E-10)	(0.0052)	(0.0327)
50	-1.00	-0.0020	0.0018	-0.0009	-1.50E-05	-1.00E+04	0.0101	0.0180	0.0195	0.0200	0.0219	0.0306
		(0.1906)	(0.2734)	(0.0112)	(0.0843)	(3.15E-06)	(0.0057)	(0.0016)	(0.0009)	(1.31E-11)	(0.0015)	(0.0086)
	-0.75	-0.0034	-1.10E+05	-9.25E+04	5.64E+04	-1.87E+03	0.0096	0.0182	0.0196	0.0200	0.0218	0.0298
		(0.1898)	(3.46E+06)	(2.92E+06)	(1.78E+06)	(5.90E+04)	(0.0059)	(0.0017)	(0.0012)	(0.0006)	(0.0017)	(0.0130)
	-0.50	-0.0730	-1.09E+12	-1.09E+12	4.96E+11	-1.45E+11	0.0079	0.0169	0.0182	0.0185	0.0202	0.0269
		(0.3283)	(2.26E+13)	(2.49E+13)	(1.05E+13)	(3.10E+12)	(0.0062)	(0.0051)	(0.0053)	(0.0053)	(0.0060)	(0.0102)
	-0.25	-0.3466	-3.81E+12	-2.51E+12	1.38E+12	-2.32E+11	0.0054	0.0122	0.0130	0.0131	0.0141	0.0195
		(0.5162)	(5.27E+13)	(3.10E+13)	(1.57E+13)	(3.10E+12)	(0.0064)	(0.0089)	(0.0095)	(0.0095)	(0.0103)	(0.0214)
	0.00	-0.0412	-0.2243	-1.1852	1.0296	-1.2787	0.0083	0.0181	0.0192	0.0192	0.0207	0.0257
		(0.2638)	(16.4336)	(12.3150)	(7.5537)	(10.2533)	(0.0065)	(0.0038)	(0.0039)	(0.0038)	(0.0043)	(0.0076)
	0.25	-0.0267	0.0396	-0.0063	-0.0049	-0.8422	0.0082	0.0183	0.0196	0.0200	0.0216	0.0304
		(0.2910)	(0.2994)	(0.0276)	(0.0613)	(0.1181)	(0.0067)	(0.0025)	(0.0017)	(6.68E-11)	(0.0015)	(0.0228)
	0.50	-0.0380	0.0395	-0.0088	-0.0029	-0.7349	0.0081	0.0180	0.0194	0.0200	0.0217	0.0313
		(0.3263)	(0.3666)	(0.0354)	(0.0601)	(0.2089)	(0.0069)	(0.0032)	(0.0021)	(3.04E-11)	(0.0016)	(0.0253)
	0.75	-0.0427	0.0445	-0.0101	-0.0015	-0.6584	0.0176	0.0194	0.0200	0.0220	0.0307	0.0070
		(0.3374)	(0.4073)	(0.0435)	(0.0626)	(0.2283)	(0.0070)	(0.0037)	(0.0020)	(1.71E-11)	(0.0021)	(0.0197)
	1.00	-0.0565	0.0677	-0.0158	-0.0019	-0.6172	0.0172	0.0194	0.0200	0.0225	0.0310	0.0071
		(0.3611)	(0.4643)	(0.0696)	(0.0629)	(0.2600)	(0.0071)	(0.0042)	(0.0019)	(8.12E-11)	(0.0028)	(0.0174)

Note: Results are based on 1,000 Monte Carlo replications. The data-generating process follows a t distribution with 5 degrees of freedom and mean zero. Empirical standard deviations are reported in parentheses. E denotes scientific notation. ν denotes the degrees of freedom, n the number of observations, and γ the power parameter. $\hat{\theta}$ is the estimate of the structural parameter θ. $\lambda_1, \lambda_2, \lambda_3$ are the Lagrange multipliers associated with the moment conditions, and δ is the Lagrange multiplier associated with the probability constraint. $\boldsymbol{\pi}$ denotes the estimated weights on the observations. Q1 and Q3 denote the first and third quartiles, respectively.

Table 4: Simulation results (Normal)

n	γ	$\hat{\theta}$	Lagrange multipliers				$\boldsymbol{\pi}$					
			λ_1	λ_2	λ_3	δ	Minimum	Q1	Median	Mean	Q3	Maximum
25	-1.00	-0.0180	-0.0522	-0.0309	0.0812	-1.00E+04	0.0270	0.0339	0.0376	0.0400	0.0447	0.0588
		(0.2317)	(0.6503)	(0.3998)	(0.7750)	(1.16E-05)	(0.0096)	(0.0049)	(0.0038)	(3.79E-09)	(0.0035)	(0.0295)
	-0.75	-0.0416	-2.82E+09	-2.47E+09	1.64E+09	-1.23E+08	0.0252	0.0327	0.0363	0.0382	0.0429	0.0540
		(0.2602)	(6.35E+10)	(5.11E+10)	(3.00E+10)	(2.83E+09)	(0.0111)	(0.0082)	(0.0086)	(0.0083)	(0.0100)	(0.0314)
	-0.50	-0.2743	-1.99E+12	-2.46E+12	1.59E+12	-1.44E+11	0.0141	0.0205	0.0231	0.0247	0.0281	0.0386
		(0.4291)	(3.77E+13)	(4.33E+13)	(2.56E+13)	(2.39E+12)	(0.0140)	(0.0167)	(0.0186)	(0.0194)	(0.0225)	(0.0463)
	-0.25	-0.4780	-2.45E+12	-2.87E+12	2.31E+12	-2.09E+11	0.0075	0.0123	0.0138	0.0148	0.0163	0.0263
		(0.4192)	(1.74E+13)	(1.91E+13)	(1.64E+13)	(1.73E+12)	(0.0124)	(0.0167)	(0.0186)	(0.0193)	(0.0220)	(0.0578)
	0.00	-0.1594	8.49E+01	-1.88E+02	6.93E+01	-1.36E+02	0.0203	0.0288	0.0317	5.35E+11	0.0366	1.34E+13
		(0.3896)	(5.99E+02)	(9.78E+02)	(3.47E+02)	(8.92E+02)	(0.0146)	(0.0141)	(0.0153)	(1.69E+13)	(0.0178)	(4.23E+14)
	0.25	-0.0003	0.0042	-0.0036	-0.0017	-0.8362	0.0233	0.0349	0.0387	0.0400	0.0453	0.0529
		(0.2250)	(0.4652)	(0.0241)	(0.2268)	(0.0528)	(0.0130)	(0.0042)	(0.0029)	(6.60E-10)	(0.0038)	(0.0133)
	0.50	0.0032	0.0119	-0.0049	-0.0062	-0.7097	0.0227	0.0351	0.0388	0.0400	0.0452	0.0525
		(0.2317)	(0.4597)	(0.0264)	(0.1966)	(0.0692)	(0.0134)	(0.0044)	(0.0029)	(3.11E-10)	(0.0037)	(0.0140)
	0.75	0.0036	0.0150	-0.0033	-0.0049	-0.6222	0.0230	0.0350	0.0389	0.0400	0.0453	0.0523
		(0.2300)	(0.4732)	(0.0235)	(0.1815)	(0.0921)	(0.0136)	(0.0048)	(0.0028)	(1.63E-10)	(0.0039)	(0.0143)
	1.00	0.0018	0.0153	-0.0046	-0.0034	-0.5637	0.0221	0.0347	0.0390	0.0400	0.0456	0.0527
		(0.2423)	(0.5024)	(0.0329)	(0.1842)	(0.1226)	(0.0137)	(0.0052)	(0.0028)	(6.90E-11)	(0.0044)	(0.0150)
50	-1.00	-0.0060	-0.0155	4.60E-05	0.0059	-1.00E+04	0.0134	0.0179	0.0194	0.0200	0.0220	0.0272
		(0.1478)	(0.3186)	(0.0038)	(0.1322)	(1.95E-06)	(0.0046)	(0.0015)	(0.0009)	(9.24E-13)	(0.0013)	(0.0061)
	-0.75	-0.0071	-3.23E+06	-2.92E+06	2.01E+06	-8.57E+04	0.0130	0.0178	0.0193	0.0199	0.0219	0.0268
		(0.1602)	(5.49E+07)	(5.06E+07)	(3.50E+07)	(1.48E+06)	(0.0050)	(0.0021)	(0.0018)	(0.0017)	(0.0023)	(0.0062)
	-0.50	-0.1240	-1.57E+12	-1.34E+12	1.02E+12	-7.21E+10	0.0105	0.0151	0.0164	0.0168	0.0186	0.0231
		(0.3453)	(4.53E+13)	(3.84E+13)	(2.96E+13)	(1.97E+12)	(0.0066)	(0.0067)	(0.0072)	(0.0073)	(0.0082)	(0.0175)
	-0.25	-0.4980	-2.33E+12	-2.54E+12	1.93E+12	-1.89E+11	0.0047	0.0076	0.0083	0.0084	0.0093	0.0117
		(0.4615)	(1.85E+13)	(2.07E+13)	(1.57E+13)	(2.01E+12)	(0.0066)	(0.0090)	(0.0097)	(0.0099)	(0.0109)	(0.0135)
	0.00	-0.0493	0.5854	-12.4907	7.1143	-4.4559	0.0117	0.0172	0.0185	0.0188	0.0206	0.0241
		(0.2509)	(110.3248)	(182.4158)	(57.5511)	(150.0305)	(0.0062)	(0.0045)	(0.0047)	(0.0047)	(0.0053)	(0.0074)
	0.25	-0.0017	-0.0130	-0.0007	0.0062	-0.8165	0.0120	0.0183	0.0197	0.0200	0.0219	0.0253
		(0.1491)	(0.2887)	(0.0077)	(0.1187)	(0.0243)	(0.0062)	(0.0014)	(0.0007)	(9.23E-11)	(0.0014)	(0.0044)
	0.50	-0.0021	-0.0094	-0.0028	0.0050	-0.6869	0.0117	0.0183	0.0198	0.0200	0.0219	0.0251
		(0.1480)	(0.2914)	(0.0162)	(0.1112)	(0.0342)	(0.0065)	(0.0016)	(0.0008)	(4.47E-11)	(0.0015)	(0.0045)
	0.75	0.0013	-0.0018	-0.0042	0.0019	-0.5990	0.0114	0.0182	0.0197	0.0200	0.0220	0.0252
		(0.1563)	(0.3178)	(0.0249)	(0.1080)	(0.0591)	(0.0067)	(0.0019)	(0.0009)	(4.64E-11)	(0.0017)	(0.0057)
	1.00	-0.0089	0.0166	-0.0063	-0.0025	-0.5346	0.0111	0.0181	0.0198	0.0200	0.0220	0.0252
		(0.1661)	(0.3275)	(0.0460)	(0.1001)	(0.0924)	(0.0068)	(0.0021)	(0.0008)	(3.36E-11)	(0.0018)	(0.0070)

Note: Results are based on 1,000 Monte Carlo replications. The data-generating process follows a normal distribution with mean zero and variance one. Empirical standard deviations are reported in parentheses. E denotes scientific notation. n the number of observations, and γ the power parameter. $\hat{\theta}$ is the estimate of the structural parameter θ. $\lambda_1, \lambda_2, \lambda_3$ are the Lagrange multipliers associated with the moment conditions, and δ is the Lagrange multiplier associated with the probability constraint. $\boldsymbol{\pi}$ denotes the estimated weights on the observations. Q1 and Q3 denote the first and third quartiles, respectively.

The influence of γ appears only in higher-order terms. A higher-order expansion yields

$$\hat{\theta} - \theta_0 = \frac{B(\gamma)}{n} + \frac{1}{\sqrt{n}} Z + o(n^{-1}),$$

which implies

$$\mathrm{MSE}_n(\gamma) = \frac{V}{n} + \frac{B(\gamma)^2}{n^2} + o(n^{-2}).$$

Hence differences in MSE across values of γ arise from second-order bias rather than from changes in asymptotic efficiency. Selecting γ to minimize MSE (or its empirical proxy via cross-validation) therefore amounts to choosing the loss geometry that minimizes finite-sample distortion while leaving first-order efficiency unchanged.

Given this interpretation, γ can be selected in a data-driven manner, for example through cross-validation. The procedure separates hyperparameter selection from structural estimation. For each candidate γ, the parameters $(\theta, \lambda, \delta)$ are estimated subject to the moment and probability constraints, while γ determines the curvature of the CRPD objective. Each candidate value of γ is then evaluated using a K-fold cross-validation criterion based on out-of-sample moment instability, which serves as a proxy for finite-sample risk. The final estimator is obtained by refitting the model on the full sample at the selected value $\hat{\gamma}$. The pseudo-algorithm is presented in Algorithm 1.

As a caveat, note that because the CRPD divergence $\mathscr{I}_\gamma(\cdot)$ varies with γ, its numerical values are not directly comparable across candidate values of γ. Each value of γ defines a different divergence and therefore a different loss geometry. Consequently, the minimized value of the CRPD objective reflects the fit of the model under that particular loss function rather than a common scale across specifications. Comparing these minimized values across γ would therefore be misleading, as the underlying optimization problems differ. Cross-validation therefore evaluates γ using a common out-of-sample moment criterion, selecting the value that yields the most stable moment fit.

6 Empirical Application

To illustrate our proposed method in a simple empirical setting, we use the dataset from Owen (2001, Table 3.2, p. 44), which reports the pounds of milk produced and the number of days milked in 1936 for $n = 22$ dairy cows. Suppose we want to estimate the mean daily milk yield θ.

Table 5 reports descriptive statistics for milk production per day (mpd). The sample consists of 22 observations. The average milk production per day is 12.43 units, with a standard deviation of 3.08, indicating moderate dispersion across cows. The minimum and maximum values are 7.55 and 18.66, respectively, suggesting substantial variation in daily milk productivity within the sample. Overall, the distribution of mpd reflects heterogeneity in production levels across cows.

Algorithm 1 CRPD Estimation with Cross-Validated Hyperparameter γ

Require: Data $\{Z_i\}_{i=1}^n$, moment function $m(Z, \theta)$, candidate set Γ, folds K
Ensure: Selected $\hat{\gamma}$ and estimates $(\hat{\theta}, \hat{\lambda}, \hat{\delta})$

Randomly partition $\{1, \dots, n\}$ into K folds $\{\mathcal{F}_k\}_{k=1}^K$.
for $\gamma \in \Gamma$ **do**
    Set $\mathrm{CV}(\gamma) \leftarrow 0$.
    for $k = 1, \dots, K$ **do**
        Training set $\mathcal{T}_k = \{1, \dots, n\} \setminus \mathcal{F}_k$; validation set $\mathcal{V}_k = \mathcal{F}_k$.
        Fit CRPD on $\mathcal{T}_k$ at fixed γ:

$$(\hat{\theta}_k(\gamma), \hat{\lambda}_k(\gamma), \hat{\delta}_k(\gamma)) \in \arg\min_{\theta,\lambda,\delta} \mathscr{I}_\gamma\big(\pi(\lambda, \delta, \theta), \mathbf{i}/|\mathcal{T}_k|\big)$$

subject to

$$\sum_{i\in\mathcal{T}_k} \pi_i = 1, \qquad \sum_{i\in\mathcal{T}_k} \pi_i \, m(Z_i, \theta) = 0.$$

        Compute out-of-sample moment instability on $\mathcal{V}_k$:

$$\mathcal{V}_k(\gamma) = \left\| \frac{1}{|\mathcal{V}_k|} \sum_{i\in\mathcal{V}_k} m\big(Z_i, \hat{\theta}_k(\gamma)\big) \right\|^2 .$$

        Update $\mathrm{CV}(\gamma) \leftarrow \mathrm{CV}(\gamma) + \mathcal{V}_k(\gamma)/K$.
    end for
end for
Select $\hat{\gamma} \in \arg\min_{\gamma\in\Gamma} \mathrm{CV}(\gamma)$.
Refit on the full sample at $\hat{\gamma}$:

$$(\hat{\theta}, \hat{\lambda}, \hat{\delta}) \in \arg\min_{\theta,\lambda,\delta} \mathscr{I}_{\hat{\gamma}}\big(\pi(\lambda, \delta, \theta), \mathbf{i}/n\big)$$

subject to

$$\sum_{i=1}^{n} \pi_i = 1, \qquad \sum_{i=1}^{n} \pi_i \, m(Z_i, \theta) = 0.$$

Table 5: Descriptive statistics for teh milk per day

Statistic	n	Mean	St. Dev.	Minimum	Maximum
mpd	22	12.4250	3.0750	7.5470	18.6610

Note: mpd denotes milk per day (lb/day). n is the number of observations. St. Dev. denotes the standard deviation.

Empirical specification. Let $X_i = mpd_i$ denote milk production per day for cow i. Suppose our parameter of interest is the population mean $\mu = \mathrm{E}[X_i]$. The identifying moment condition is therefore $\mathrm{E}[X_i - \mu] = 0$. However, with a single moment condition the CRPD estimator collapses to the trivial uniform-weight solution, since the sample mean satisfies the moment restriction exactly.

To generate an overidentified system, we introduce an instrument moment based on the observable variable $Days_i$. Specifically, we impose $\mathrm{E}\big[(X_i - \mu)\, Days_i\big] = 0$. This restriction implies that deviations of daily milk production from the population mean should not systematically vary with the number of observed days. A Pearson correlation test yields a p-value of 0.32, indicating that the correlation is not statistically significant and that the moment condition is consistent with the data. Thus, we consider the moment system

$$g(X_i, \mu) = \begin{pmatrix} X_i - \mu \\ (X_i - \mu)Days_i \end{pmatrix},$$

which provides two moment conditions for the single parameter μ, yielding an overidentified model ($q = 2 > p = 1$) suitable for CRPD estimation.

Data-driven selection of γ. The CRPD estimator depends on the power parameter γ, which determines the curvature of the divergence and therefore the implied weighting scheme. To select γ in a data-driven way, we implement five-fold cross-validation over a grid $\gamma \in [-10, 10]$ with step size 0.001. For each candidate value of γ, the estimator is computed on the training folds, and the out-of-sample risk is evaluated on the validation fold using the squared prediction loss $\frac{1}{n_v}\sum_{i\in V}(X_i - \hat{\mu})^2$, where V denotes the validation sample and n_v its size. The cross-validated estimate $\hat{\gamma}$ is chosen as the value minimizing the average validation risk across folds. The final CRPD estimator of μ is then obtained by re-estimating the model using the full sample at $\gamma = \hat{\gamma}$.

Estimation results. Table 6 reports the estimation results across several specifications of the CRPD power parameter. When γ is selected via cross-validation, the procedure yields $\hat{\gamma} = -0.654$ and $\hat{\theta} = 12.8245$ lb/day. The table also illustrates the potential risks of selecting a specific value of γ a priori: extreme values of the power parameter can substantially alter the estimator. For example, when $\gamma = -1.5$ or $\gamma = -2$, the estimated mean shifts noticeably and the divergence criterion increases sharply. This behavior indicates that highly negative values of γ produce a loss geometry that requires substantial distortion of the empirical distribution in order to satisfy the moment conditions. In contrast, for moderate values of γ the estimator remains stable, suggesting that the data are broadly compatible with the moment restrictions without excessive reweighting.

While the point estimate $\hat{\theta}$ remains nearly invariant across moderate values of γ up to the fourth decimal place, the implied probability weights $\hat{\boldsymbol{\pi}}(\gamma)$ differ even when $\hat{\theta}$ values appear similar (for example, between the pre-specified choices $\gamma = -1$, $\gamma = 0$, and the cross-validated $\hat{\gamma} = -0.654$). Figure 2 illustrates these differences in the estimated weights. The dashed line denotes the uniform weight benchmark $1/n$. When γ is close to zero (ET), the weights cluster tightly around this benchmark, indicating that the moment conditions are nearly satisfied under equal weighting. As γ becomes more negative (EL-like), the weight distribution becomes more dispersed, assigning greater influence to a subset of observations in order to balance the moment equations. The cross-validated estimate $\hat{\gamma} = -0.654$ lies between these regimes and produces an intermediate weighting scheme.

Notably, the estimated probability weights $\hat{\boldsymbol{\pi}}(\gamma) = \{\hat{\pi}_i(\gamma)\}_{i=1}^n$ provide useful diagnostic information about how the estimator uses the data. Because the estimator can be written as $\hat{\theta}(\gamma) = \sum_{i=1}^n \hat{\pi}_i(\gamma) X_i$, the distribution of the weights reveals how influence is allocated across observations. Weights close to the uniform benchmark $1/n$ indicate that the moment restrictions are broadly compatible with the empirical distribution and that the estimator effectively averages information across the sample. In contrast, a more dispersed or heavy-tailed weight distribution indicates that the estimator relies disproportionately on certain observations in order to satisfy the moment conditions.

Such patterns provide practical insight into the structure of the data: unusually large weights may highlight observations that carry strong identifying information or exert substantial influence on the moment equations, while systematic deviations from uniform weighting may signal heterogeneity or tension between the model restrictions and the observed sample. In some cases, clusters of observations with similar weights may suggest latent subgroup structure, indicating that the global moment restriction interacts differently with different parts of the data.

Overall, the results illustrate why γ should be interpreted as a continuum parameter and selected in a data-driven manner. Different values of γ correspond to different ways of reconciling the sample with the imposed moment restrictions through reweighting of the empirical distribution. Even when the point estimate $\hat{\theta}$ appears insensitive to γ, the distribution of the estimated weights can differ substantially, revealing meaningful differences in the implied weighting schemes and hence in the finite-sample geometry of the estimator. Selecting γ adaptively helps avoid extreme weighting schemes that may introduce non-negligible second-order bias in $\hat{\theta}$ or lead to disproportionate influence of a small subset of observations.

Table 6: Estimates for $\theta(=\mu)$ for different values of γ

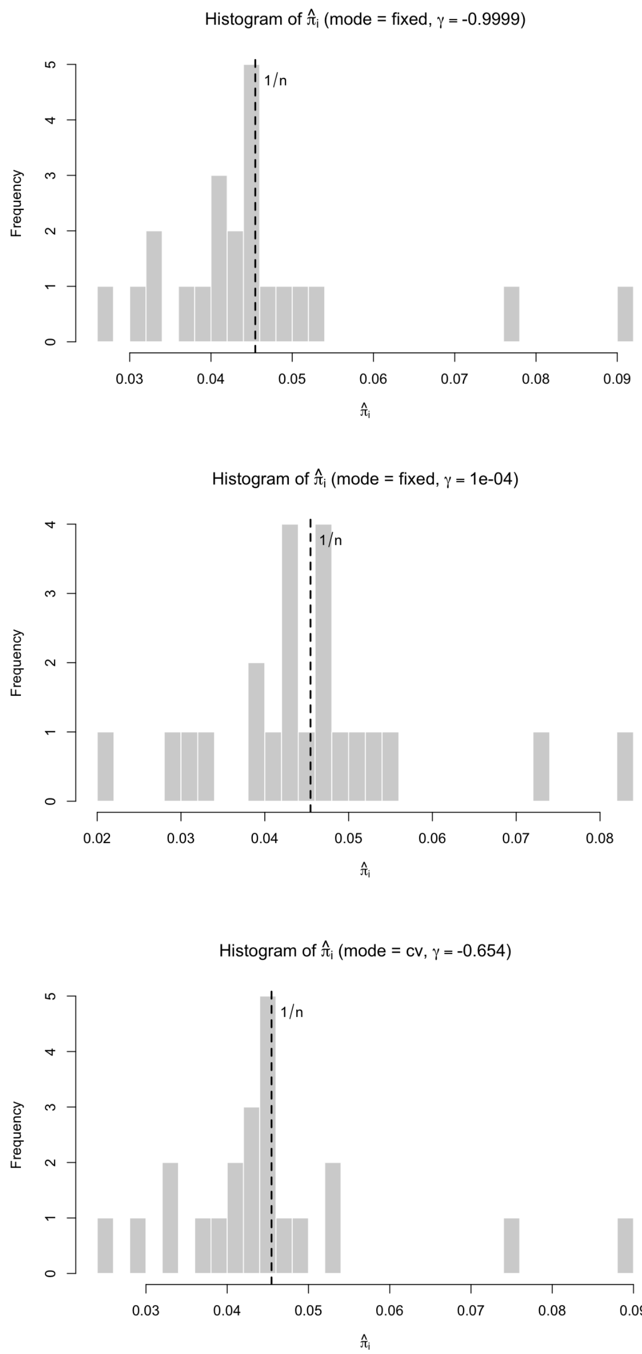
Mode	γ	$\hat{\theta}$
Pre-specified	-2	11.5790
Pre-specified	-1.5	11.0546
Pre-specified	-1	12.8245
Pre-specified	-0.5	12.8245
Pre-specified	0	12.8245
Pre-specified	0.5	12.8245
Pre-specified	1	12.8245
Cross-validated	-0.654	12.8245

Note: γ denotes the CRPD power parameter. $\hat{\theta}$ is the estimated population mean of daily milk production (lb/day). CRPD reports the minimized value of the CRPD divergence criterion. "Pre-specified" refers to specifications in which γ is fixed a priori, while "Cross-validated" refers to the value of γ selected by minimizing the out-of-sample mean squared prediction error via cross-validation.

7 Conclusion

This paper develops the Cressie–Read power divergence (CRPD) framework for moment-based estimation with a focus on finite-sample behavior, interpreting the power pa-

Figure 2: Estimated observation weights $\hat{\boldsymbol{\pi}}$ for alternative choices of γ



Note: The estimated value of γ using our method is -0.654 (reported in the third row), while the remaining rows correspond to specifications in which γ is fixed a priori at -1 (EL) and 0 (ET).

rameter γ as a hyperparameter. The CRPD family is indexed by γ, which has traditionally been treated as a researcher-chosen tuning index, with common benchmark values including empirical likelihood ($\gamma = -1$) and exponential tilting ($\gamma = 0$). We argue that γ is more naturally understood as a parameter governing the curvature of the divergence objective. Through this channel, it shapes the geometry of the implied weighting scheme and can materially affect finite-sample performance, even though it does not alter population identification.

This interpretation clarifies how divergence-based moment estimation can be adapted to the empirical environment. The choice of γ controls how the estimator reweights observations in response to moment residuals, thereby affecting robustness and sensitivity to sampling variation. At the same time, the moment-based framework preserves structural transparency: the economic content remains encoded in explicitly stated moment restrictions, ensuring interpretability of the resulting estimates. Treating γ as a hyperparameter thus provides a systematic way to tailor finite-sample behavior while maintaining the discipline of the underlying econometric model.

An important direction for future research is to study CRPD estimation and the role of γ under moment misspecification. Under misspecification, the first-order equivalence across members of the CRPD class generally breaks down (see, e.g., Schennach (2007)), so γ may influence not only higher-order behavior but also the leading asymptotic approximation. Characterizing the pseudo-true targets, robustness properties, and optimal data-driven selection of γ in misspecified environments, and their implications for inference, would further strengthen the ML interpretation of CRPD.

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References

Cressie, N. and Read, T. R. (1984). Multinomial goodness-of-fit tests. *Journal of the Royal Statistical Society Series B: Statistical Methodology*, 46(3):440–464.

Hansen, L., Heaton, J., and Yaron, A. (1996). Finite-sample properties of some tive gmm estimators. *Journal of Business and Economic Statistics*, 14:262–280.

Huber, P. (1964). Robust estimation of a location parameter. *The Annals of Mathematical Statistics*, 35(1):73–101.

Huber, P. (1973). Robust regression: asymptotics, conjectures and monte carlo. *The Annals of Statistics*, pages 799–821.

Huber, P. and Ronchetti, E. (2009). *Robust Statistics*. Wiley.

Imbens, G. W., Spady, R. H., and Johnson, P. (1998). Information theoretic approaches to inference in moment conditions models. *Econometrica*, 66(2):333–357.

Kitamura, Y. and Stutzer, M. (1997). An information-theoretic alternative to generalized method of moments estimation. *Econometrica*, 65:861–874.

Newey, W. K. and Smith, R. J. (2004). Higher order properties of gmm and generalized empirical likelihood estimators. *Econometrica*, 72(1):219–255.

Owen, A. B. (1988). Empirical likelihood ratio confidence intervals for a single functional. *Biometrika*, 75(2):237–249.

Qin, J. and Lawless, J. (1994). Empirical likelihood and general estimating equations. *The Annals of Statistics*, 22(1):300–3235.

Rothenberg, T. J. (1984). *Handbook of econometrics (Approximating the distributions of econometric estimators and test statistics)*, volume 2.

Schennach, S. (2007). Point estimation with exponentially tilted empirical likelihood. *The Annals of Statistics*, 35(2):634–672.

Serfling, R. J. (1980). *Approximation theorems of mathematical statistics*. John Wiley & Sons.

Toulis, P. and Airoldi, E. M. (2017). Asymptotic and finite-sample properties of estimators based on stochastic gradients. *Annals of Statistics*, 45(4):1694–1727.

Appendix

A.1 Proof for Lemma 3.1

To characterize the population solution, consider the population CRPD problem obtained by replacing sample averages with expectations: $\min_P \mathscr{I}_\gamma(P, P_0)$ subject to $\mathbb{E}_P[g(X,\theta)] = 0$, where P_0 denotes the true data-generating distribution.

Under correct specification, $\mathbb{E}_{P_0}[g(X,\theta_0)] = 0$, so the true distribution P_0 already satisfies the moment restriction. Since CRPD divergences are uniquely minimized at the reference distribution, the population minimizer is $P = P_0,\ \lambda_0 = 0$, that is, no distortion of the distribution is required when the moment condition holds exactly.

At the sample level, the empirical distribution with uniform weights $\pi_i = 1/n$ is the natural analogue of P_0. By the Glivenko–Cantelli theorem, the empirical distribution converges uniformly to P_0. Hence, consistency of the CRPD estimator implies $\hat{\lambda} \xrightarrow{p} 0$.

More explicitly, under correct specification the population moment condition satisfies $\mathbb{E}[g(X,\theta_0)] = 0$. By the Law of Large Numbers, $\frac{1}{n}\sum_{i=1}^n g_i(\theta_0) = O_p(n^{-1/2})$, so the sample moment violation at $\theta = \theta_0$ is asymptotically negligible.

Substituting $\pi_i = 1/n$ into (4) yields $\frac{1}{\gamma+1} - \gamma\delta_0 = 1$, which implies $\delta_0 = -\frac{1}{\gamma+1}$. Therefore, under correct specification the solution lies in a neighborhood of
$(\theta_0, \lambda_0, \pi_i, \delta_0) = \left(\theta_0, 0, \frac{1}{n}, -\frac{1}{\gamma+1}\right)$, as claimed. □

A.2 Proof for Theorem 3.1

Step 1: Uniform LLN for the multiplier system.

By Assumptions 3.1 and 3.2, Θ is compact and $g(z,\theta)$ is continuously differentiable with square-integrable envelope. By Assumption 3.4, with probability approaching one, there exists a compact set $\mathcal{H} = \left\{(\lambda,\delta) : \|\lambda\| + \|\delta - \delta_0\| \le \kappa\right\}$ such that for all $(\theta,\lambda,\delta) \in \mathcal{N} \times \mathcal{H}$, $\frac{1}{\gamma+1} - \gamma\delta - \gamma\lambda' g_i(\theta) \ge \kappa > 0$. On this interior region, $w_i(\theta,\lambda,\delta)$ is continuously differentiable in (θ,λ,δ) and, by the positivity bound in Assumption 3.4 together with Assumption 3.2, is uniformly bounded by an integrable envelope function. Consequently, the empirical averages of $w(Z;\theta,\lambda,\delta)$ and $w(Z;\theta,\lambda,\delta)g(Z,\theta)$ satisfy a uniform law of large numbers over $(\theta,\lambda,\delta) \in \mathcal{N} \times \mathcal{H}$; that is, $\sup_{(\theta,\lambda,\delta)\in\mathcal{N}\times\mathcal{H}} \left|\frac{1}{n}\sum_{i=1}^n w_i(\theta,\lambda,\delta) - E[w(Z;\theta,\lambda,\delta)]\right| \xrightarrow{p} 0$, and
$\sup_{(\theta,\lambda,\delta)\in\mathcal{N}\times\mathcal{H}} \left\|\frac{1}{n}\sum_{i=1}^n w_i(\theta,\lambda,\delta) g_i(\theta) - E[w(Z;\theta,\lambda,\delta)g(Z,\theta)]\right\| \xrightarrow{p} 0$.
Therefore, $\sup_{(\theta,\lambda,\delta)\in\mathcal{N}\times\mathcal{H}} \left\|\Psi_n(\theta,\lambda,\delta) - \Psi(\theta,\lambda,\delta)\right\| \xrightarrow{p} 0$.

Step 2: Uniform convergence of profiled multipliers.

By Assumption 3, the Jacobian $\partial_{(\lambda,\delta)}\Psi(\theta_0, \lambda_0, \delta_0)$ is nonsingular. By continuity, it remains nonsingular on a neighborhood $\mathcal{N}$ of θ_0. Combining the uniform convergence of Ψ_n with the nonsingularity of the Jacobian and standard uniform Z-estimation arguments,

$$\sup_{\theta\in\mathcal{N}} \left\| \begin{pmatrix} \hat{\lambda}(\theta) \\ \hat{\delta}(\theta) \end{pmatrix} - \begin{pmatrix} \lambda(\theta) \\ \delta(\theta) \end{pmatrix} \right\| \xrightarrow{p} 0.$$

Step 3: Uniform convergence of the profiled objective.

On the interior region, the map $(\theta, \lambda, \delta) \mapsto \mathscr{I}_\gamma\big(\pi(\theta, \lambda, \delta), i/n\big)$ is continuous and uniformly continuous on $\mathcal{N} \times \mathcal{H}$. Hence, $\sup_{\theta\in\mathcal{N}}\big|L_n(\theta) - L(\theta)\big| \xrightarrow{p} 0$.

Step 4: Identification and uniqueness.

By Assumption 3.3, $E[g(Z, \theta)] = 0$ if and only if $\theta = \theta_0$. Under correct specification, the CRPD divergence is minimized only when the implied weights satisfy the moment restriction, which occurs uniquely at θ_0. Therefore, $L(\theta)$ is uniquely minimized at θ_0. □

A.3 Proof for Lemma 3.2

Fix $\theta \in \mathcal{N}$. By Assumption 3.4, there exists an interior solution in the domain where all terms $s_i(\theta, \lambda, \delta) := \frac{1}{\gamma+1} - \gamma\delta - \gamma\lambda' g_i(\theta)$ are bounded away from zero, ensuring that $w_i = s_i^{1/\gamma}$ is well-defined and smooth.

Consider the mapping $(\lambda, \delta) \mapsto \Psi_n(\theta, \lambda, \delta)$ on the compact set $\{(\lambda, \delta) : \|\lambda\| + \|\delta - \delta_0\| \le \kappa\}$ intersected with the positivity domain $s_i \ge \kappa$. On this set, Ψ_n is continuously differentiable. Compute the Jacobian with respect to (λ, δ):

$$\frac{\partial w_i}{\partial \lambda} = -\Big(\frac{1}{\gamma+1} - \gamma\delta - \gamma\lambda' g_i(\theta)\Big)^{1/\gamma-1} g_i(\theta), \qquad \frac{\partial w_i}{\partial \delta} = -\Big(\frac{1}{\gamma+1} - \gamma\delta - \gamma\lambda' g_i(\theta)\Big)^{1/\gamma-1}$$

Thus,

$$\frac{\partial \Psi_{n,1}}{\partial \lambda'} = -\frac{1}{n}\sum_{i=1}^{n} s_i^{1/\gamma-1} g_i(\theta)', \qquad \frac{\partial \Psi_{n,1}}{\partial \delta} = -\frac{1}{n}\sum_{i=1}^{n} s_i^{1/\gamma-1},$$

and

$$\frac{\partial \Psi_{n,2}}{\partial \lambda'} = -\frac{1}{n}\sum_{i=1}^{n} s_i^{1/\gamma-1} g_i(\theta) g_i(\theta)', \; \frac{\partial \Psi_{n,2}}{\partial \delta} = -\frac{1}{n}\sum_{i=1}^{n} s_i^{1/\gamma-1} g_i(\theta).$$

At $(\lambda, \delta) = (0, \delta_0)$ we have $s_i = 1$ and hence $\frac{\partial \Psi_{n,1}}{\partial \lambda'}\Big|_{(0,\delta_0)} = -\bar{g}(\theta)'$, $\frac{\partial \Psi_{n,1}}{\partial \delta}\Big|_{(0,\delta_0)} = -1$,

$\frac{\partial \Psi_{n,2}}{\partial \lambda'}\Big|_{(0,\delta_0)} = -\hat{\Omega}(\theta), \qquad \frac{\partial \Psi_{n,2}}{\partial \delta}\Big|_{(0,\delta_0)} = -\bar{g}(\theta)$, where $\bar{g}(\theta) := \frac{1}{n}\sum_{i=1}^n g_i(\theta)$ and $\hat{\Omega}(\theta) := \frac{1}{n}\sum_{i=1}^n g_i(\theta)g_i(\theta)'$. Denote the Jacobian with respect to (λ', δ) at $(\lambda, \delta) = (0, \delta_0)$ by

$$J_n(\theta) := \frac{\partial \Psi_n(\theta, \lambda, \delta)}{\partial(\lambda', \delta)}\Bigg|_{(\lambda,\delta)=(0,\delta_0)} = \begin{pmatrix} -\bar{g}(\theta)' & -1 \\ -\hat{\Omega}(\theta) & -\bar{g}(\theta) \end{pmatrix}.$$

Since signs do not affect nonsingularity, it suffices to consider $\tilde{J}_n(\theta) = \begin{pmatrix} \bar{g}(\theta)' & 1 \\ \hat{\Omega}(\theta) & \bar{g}(\theta) \end{pmatrix}$. Permute rows and columns to place the scalar block first. Let P be the permutation matrix that swaps the $(q+1)$st coordinate with the first q coordinates, so that $P\,\tilde{J}_n(\theta)\,P' = \begin{pmatrix} 1 & \bar{g}(\theta)' \\ \bar{g}(\theta) & \hat{\Omega}(\theta) \end{pmatrix}$. Because permutation preserves determinant up to sign, $\tilde{J}_n(\theta)$ is nonsingular if and only if $P\tilde{J}_n(\theta)P'$ is nonsingular. By Assumption 3.3 and continuity, $\hat{\Omega}(\theta)$ is nonsingular w.p.a.1 for θ near θ_0. Using the Schur complement with respect to $\hat{\Omega}(\theta)$ yields $\det\big(P\tilde{J}_n(\theta)P'\big) = \det\big(\hat{\Omega}(\theta)\big)\left(1 - \bar{g}(\theta)'\hat{\Omega}(\theta)^{-1}\bar{g}(\theta)\right)$. Since $\hat{\Omega}(\theta)$ is positive definite w.p.a.1, the quadratic form $\bar{g}(\theta)'\hat{\Omega}(\theta)^{-1}\bar{g}(\theta) \geq 0$. Moreover, for θ in a shrinking neighborhood of θ_0, $\bar{g}(\theta) = O_p(n^{-1/2})$, so $\bar{g}(\theta)'\hat{\Omega}(\theta)^{-1}\bar{g}(\theta) = O_p(n^{-1})$, and hence $1 - \bar{g}(\theta)'\hat{\Omega}(\theta)^{-1}\bar{g}(\theta) = 1 + o_p(1)$. Therefore the Schur complement is strictly positive w.p.a.1 for n large, implying that $J_n(\theta)$ is nonsingular in a neighborhood of $(0, \delta_0)$ w.p.a.1.

Therefore, by the Implicit Function Theorem there exists a neighborhood $\mathcal{N}$ of θ_0 and a unique continuously differentiable mapping $\theta \mapsto \big(\lambda(\theta), \delta(\theta)\big)$ defined on $\mathcal{N}$ such that $\Psi_n\big(\theta, \lambda(\theta), \delta(\theta)\big) = 0$ for all $\theta \in \mathcal{N}$. In particular, the solution (λ, δ) is locally unique in a neighborhood of $(0, \delta_0)$, and depends smoothly on θ. □

A.4 Proof for Theorem 3.2

By Theorem 3.1, the profiled criterion $L_n(\theta)$ converges uniformly in probability to a deterministic limit function $L(\theta)$ on Θ, that is, $\sup_{\theta\in\Theta} |L_n(\theta) - L(\theta)| \xrightarrow{p} 0$. By Assumption 3.3, the population criterion $L(\theta)$ is uniquely minimized at θ_0. Under Assumption 3.1, the conditions of the standard argmin theorem for M-estimators are satisfied. Therefore, any measurable selection $\hat{\theta} \in \arg\min_{\theta\in\Theta} L_n(\theta)$ satisfies $\hat{\theta} \xrightarrow{p} \theta_0$. □

A.5 Proof for Corollary 3.1

By Theorem 3.2, $\hat{\theta} \xrightarrow{p} \theta_0$. Lemma 3.2 implies the mapping $\theta \mapsto (\lambda(\theta), \delta(\theta))$ is continuous at θ_0, the Continuous Mapping Theorem implies $(\lambda(\hat{\theta}), \delta(\hat{\theta})) \xrightarrow{p} (\lambda(\theta_0), \delta(\theta_0))$. By definition, $(\hat{\lambda}, \hat{\delta}) = (\lambda(\hat{\theta}), \delta(\hat{\theta}))$, which yields the result. $\square$

A.6 Proof for Theorem 3.3

Because $(\hat{\lambda}(\hat{\theta}), \hat{\delta}(\hat{\theta}))$ satisfies $\Psi_n(\hat{\theta}, \hat{\lambda}(\hat{\theta}), \hat{\delta}(\hat{\theta})) = 0$ and $\hat{\theta} \xrightarrow{p} \theta_0$, it suffices to expand $\Psi_n(\theta, \lambda(\theta), \delta(\theta))$ in $(\lambda(\theta), \delta(\theta))$ around $(0, \delta_0)$ with θ in a shrinking neighborhood of θ_0. For notational simplicity, we suppress the explicit profiling dependence on θ in what follows and write $(\hat{\lambda}, \hat{\delta})$ for $(\hat{\lambda}(\hat{\theta}), \hat{\delta}(\hat{\theta}))$, and (λ, δ) for $(\lambda(\theta), \delta(\theta))$ whenever no confusion arises.

Write $s_i(\theta, \lambda, \delta) = \frac{1}{\gamma+1} - \gamma\delta - \gamma\lambda' g_i(\theta)$ and $w_i(\theta, \lambda, \delta) = s_i(\theta, \lambda, \delta)^{1/\gamma}$. By Lemma 3.2, with probability approaching one, $(\hat{\lambda}, \hat{\delta})$ lies in a neighborhood of $(0, \delta_0)$ on which $\inf_i s_i(\hat{\theta}, \lambda, \delta) \geq c > 0$. Hence $w_i(\theta, \lambda, \delta)$ is twice continuously differentiable in (λ, δ) on that neighborhood, and its first derivatives are Lipschitz (uniformly in i) on the same set.

Step 1: Mean-value expansion and remainder bound. Apply a mean-value expansion in (λ, δ) around $(0, \delta_0)$:

$$w_i(\hat{\theta}, \hat{\lambda}, \hat{\delta}) = w_i(\hat{\theta}, 0, \delta_0) + \frac{\partial w_i}{\partial \delta}\Big|_{(\hat{\theta}, \tilde{\lambda}, \tilde{\delta})} (\hat{\delta} - \delta_0) + \frac{\partial w_i}{\partial \lambda'}\Big|_{(\hat{\theta}, \tilde{\lambda}, \tilde{\delta})} \hat{\lambda},$$

for some intermediate $(\tilde{\lambda}, \tilde{\delta})$ on the line segment between $(0, \delta_0)$ and $(\hat{\lambda}, \hat{\delta})$. Since $w_i(\hat{\theta}, 0, \delta_0) = 1$ by the choice of δ_0, define the remainder

$$r_{i,n} := \Big(\frac{\partial w_i}{\partial \delta}\Big|_{(\hat{\theta}, \tilde{\lambda}, \tilde{\delta})} - \frac{\partial w_i}{\partial \delta}\Big|_{(\hat{\theta}, 0, \delta_0)} \Big) (\hat{\delta} - \delta_0) + \Big(\frac{\partial w_i}{\partial \lambda'}\Big|_{(\hat{\theta}, \tilde{\lambda}, \tilde{\delta})} - \frac{\partial w_i}{\partial \lambda'}\Big|_{(\hat{\theta}, 0, \delta_0)} \Big) \hat{\lambda}.$$

Because the derivatives are Lipschitz on the neighborhood (uniformly in i),

$$\max_{1 \leq i \leq n} |r_{i,n}| = O_p\big(\|\hat{\lambda}\|^2 + |\hat{\delta} - \delta_0|^2 \big). \tag{A.1}$$

Moreover, by smoothness in θ and $\hat{\theta} \to_p \theta_0$,

$$\frac{\partial w_i}{\partial \delta}\Big|_{(\hat{\theta},0,\delta_0)} = -1 + o_p(1), \qquad \frac{\partial w_i}{\partial \lambda'}\Big|_{(\hat{\theta},0,\delta_0)} = -g_i(\hat{\theta})' + o_p(1),$$

uniformly in i on events of probability approaching one. Therefore,

$$w_i(\hat{\theta}, \hat{\lambda}, \hat{\delta}) = 1 - (\hat{\delta} - \delta_0) - \hat{\lambda}' g_i(\hat{\theta}) + \rho_{i,n} + r_{i,n}, \tag{A.2}$$

where $\max_i |\rho_{i,n}| = o_p(1)\big(\|\hat{\lambda}\| + |\hat{\delta} - \delta_0|\big)$ and $r_{i,n}$ satisfies (A.1).

Step 2: Moment constraint yields the rate and influence function for $\hat{\lambda}$. Next plug (A.2) into $\Psi_{n,2}(\hat{\theta}, \hat{\lambda}, \hat{\delta}) = 0$:

$$\begin{aligned} 0 &= \frac{1}{n}\sum_{i=1}^{n} w_i(\hat{\theta}, \hat{\lambda}, \hat{\delta})\, g_i(\hat{\theta}) \\ &= \bar{g}(\hat{\theta}) - (\hat{\delta} - \delta_0)\bar{g}(\hat{\theta}) - \hat{\Omega}(\hat{\theta})\hat{\lambda} + \tilde{\rho}_n + \tilde{r}_n, \end{aligned} \tag{A.3}$$

where $\hat{\Omega}(\hat{\theta}) := n^{-1}\sum_i g_i(\hat{\theta})g_i(\hat{\theta})'$, $\tilde{\rho}_n := n^{-1}\sum_i \rho_{i,n} g_i(\hat{\theta})$, and $\tilde{r}_n := n^{-1}\sum_i r_{i,n} g_i(\hat{\theta})$. Under Assumption 3.2 and the definitions above, $\tilde{\rho}_n = o_p(1)\big(\|\hat{\lambda}\| + |\hat{\delta} - \delta_0|\big)$, $\tilde{r}_n = O_p\big(\|\hat{\lambda}\|^2 + |\hat{\delta} - \delta_0|^2\big)$. Since $\bar{g}(\hat{\theta}) = O_p(n^{-1/2})$ and $\hat{\delta} - \delta_0 = o_p(1)$ by Corollary 3.1, the term $(\hat{\delta} - \delta_0)\bar{g}(\hat{\theta}) = o_p(n^{-1/2})$. Hence,

$$\hat{\Omega}(\hat{\theta})\hat{\lambda} = \bar{g}(\hat{\theta}) + o_p(n^{-1/2}) + o_p(1)\big(\|\hat{\lambda}\| + |\hat{\delta} - \delta_0|\big) + O_p\big(\|\hat{\lambda}\|^2 + |\hat{\delta} - \delta_0|^2\big). \tag{A.4}$$

Since $\hat{\Omega}(\hat{\theta}) \xrightarrow{p} \Omega_0 \succ 0$, we have $\|\hat{\Omega}(\hat{\theta})^{-1}\| = O_p(1)$. Thus, multiplying (A.4) by $\hat{\Omega}(\hat{\theta})^{-1}$ and taking norms yields

$$\|\hat{\lambda}\| \le C\Big(\|\bar{g}(\hat{\theta})\| + o_p(n^{-1/2}) + o_p(1)\big(\|\hat{\lambda}\| + |\hat{\delta} - \delta_0|\big) + O_p\big(\|\hat{\lambda}\|^2 + |\hat{\delta} - \delta_0|^2\big)\Big)$$

for some constant $C > 0$ on an event of probability approaching one.

By Lemma 3.2, with probability approaching one the solution $(\hat{\lambda}, \hat{\delta})$ lies in a fixed neighborhood of $(0, \delta_0)$, so that $\|\hat{\lambda}\| + |\hat{\delta} - \delta_0| \le \kappa$ for some $\kappa > 0$. In particular, since $x^2 \le \kappa x$ whenever $0 \le x \le \kappa$, $\|\hat{\lambda}\|^2 + |\hat{\delta} - \delta_0|^2 \le \kappa\big(\|\hat{\lambda}\| + |\hat{\delta} - \delta_0|\big)$. Hence the quadratic remainder term in (A.4) is bounded by a constant multiple of $\|\hat{\lambda}\| + |\hat{\delta} - \delta_0|$ on this neighborhood.

Combining the bounds above with (A.4) and multiplying by $\|\hat{\Omega}(\hat{\theta})^{-1}\|$, which is $O_p(1)$ since $\hat{\Omega}(\hat{\theta}) \to_p \Omega_0 \succ 0$, we obtain on an event of probability approaching one

$$\|\hat{\lambda}\| \leq C\|\bar{g}(\hat{\theta})\| + C\, o_p(1)\big(\|\hat{\lambda}\| + |\hat{\delta} - \delta_0|\big) + C\,\kappa\big(\|\hat{\lambda}\| + |\hat{\delta} - \delta_0|\big),$$

for some deterministic constant $C > 0$. Collecting the coefficients multiplying $\|\hat{\lambda}\|$ into $\eta_n := C\, o_p(1) + C\kappa$, we can rewrite the inequality as $\|\hat{\lambda}\| \leq C\|\bar{g}(\hat{\theta})\| + \eta_n\|\hat{\lambda}\| + C'|\hat{\delta} - \delta_0|$, where $C' > 0$ is another finite constant absorbing fixed multiplicative factors. Because $o_p(1) \to 0$ and κ is fixed and can be chosen sufficiently small (by restricting to the local solution), we have $\eta_n = o_p(1)$. Hence for n sufficiently large, $\eta_n < 1/2$ w.p.a.1. Rearranging gives $(1-\eta_n)\|\hat{\lambda}\| \leq C\|\bar{g}(\hat{\theta})\| + C'|\hat{\delta}-\delta_0|$, and therefore $\|\hat{\lambda}\| \leq \frac{C}{1-\eta_n}\big(\|\bar{g}(\hat{\theta})\| + |\hat{\delta} - \delta_0|\big)$. Since $\hat{\delta} - \delta_0 = o_p(1)$, $(1 - \eta_n)^{-1} = O_p(1)$ and $\bar{g}(\hat{\theta}) = O_p(n^{-1/2})$, this implies $\hat{\lambda} = O_p(n^{-1/2})$. Specifically, using $\bar{g}(\hat{\theta}) = \bar{g}(\theta_0) + G_0(\hat{\theta} - \theta_0) + o_p(n^{-1/2})$, $\hat{\Omega}(\hat{\theta}) = \hat{\Omega}(\theta_0) + o_p(1)$, we obtain the first-order representation of (A.4) $\hat{\lambda} = \hat{\Omega}(\theta_0)^{-1}\bar{g}(\theta_0) + o_p(n^{-1/2})$, so $\sqrt{n}\,\hat{\lambda} \xrightarrow{d} N(0, \Omega_0^{-1})$ by the CLT for $\sqrt{n}\,\bar{g}(\theta_0)$.

Step 3: Probability constraint yields the rate for $\hat{\delta} - \delta_0$. Plug (A.2) into $\Psi_{n,1}(\hat{\theta}, \hat{\lambda}, \hat{\delta}) = 0$:

$$0 = \frac{1}{n}\sum_{i=1}^{n}\big[1 - (\hat{\delta} - \delta_0) - \hat{\lambda}' g_i(\hat{\theta}) + \rho_{i,n} + r_{i,n}\big] - 1 = -(\hat{\delta} - \delta_0) - \hat{\lambda}'\bar{g}(\hat{\theta}) + \bar{\rho}_n + \bar{r}_n,$$

where $\bar{\rho}_n := n^{-1}\sum_i \rho_{i,n}$ and $\bar{r}_n := n^{-1}\sum_i r_{i,n}$. Hence,

$$\hat{\delta} - \delta_0 = -\hat{\lambda}'\bar{g}(\hat{\theta}) + \bar{\rho}_n + \bar{r}_n. \tag{A.5}$$

By construction, $\bar{\rho}_n = o_p(1)\big(\|\hat{\lambda}\| + |\hat{\delta} - \delta_0|\big)$, $\bar{r}_n = O_p\big(\|\hat{\lambda}\|^2 + |\hat{\delta} - \delta_0|^2\big)$. By Step 2 we have $\hat{\lambda} = O_p(n^{-1/2})$, $\bar{g}(\hat{\theta}) = O_p(n^{-1/2})$. Therefore the leading term satisfies $\hat{\lambda}'\bar{g}(\hat{\theta}) = O_p(n^{-1/2}) \cdot O_p(n^{-1/2}) = O_p(n^{-1})$. Next consider the remainder terms in (A.5): $\bar{\rho}_n = o_p(1)\big(\|\hat{\lambda}\| + |\hat{\delta} - \delta_0|\big) = o_p(n^{-1/2}) + o_p(1)\,|\hat{\delta} - \delta_0|$, and $\bar{r}_n = O_p\big(\|\hat{\lambda}\|^2 + |\hat{\delta} - \delta_0|^2\big) = O_p(n^{-1}) + O_p\big(|\hat{\delta}-\delta_0|^2\big)$. Substituting these bounds into (A.5) gives $\hat{\delta}-\delta_0 = -\hat{\lambda}'\bar{g}(\hat{\theta}) + O_p(n^{-1}) + o_p(n^{-1/2}) + o_p(1)\,|\hat{\delta} - \delta_0| + O_p\big(|\hat{\delta} - \delta_0|^2\big)$.

Since $|\hat{\delta} - \delta_0| = o_p(1)$ by Corollary 3.1, the terms $o_p(1)\,|\hat{\delta} - \delta_0|$ and $O_p(|\hat{\delta} - \delta_0|^2)$ are of smaller order than $|\hat{\delta} - \delta_0|$ itself and therefore negligible relative to the leading terms on the right-hand side. Consequently, $\hat{\delta} - \delta_0 = -\hat{\lambda}'\bar{g}(\hat{\theta}) + O_p(n^{-1}) + o_p(n^{-1/2})$, and $\hat\lambda'\bar g(\hat\theta) = O_p(n^{-1})$. Hence all terms on the right-hand side are of order $O_p(n^{-1})$, which implies $\hat\delta - \delta_0 = O_p(n^{-1})$. □

A.7 Proof for Theorem 3.4

Define $t_i(\theta) := (\hat\delta - \delta_0) + \hat\lambda' g_i(\theta)$, so that $w_i = (1-\gamma t_i)^{1/\gamma}$ because $\frac{1}{\gamma+1} - \gamma\hat\delta - \gamma\hat\lambda' g_i(\theta) = 1 - \gamma\big((\hat\delta - \delta_0) + \hat\lambda' g_i(\theta)\big)$ when $\delta_0 = -1/(\gamma+1)$. Note that at the population solution under correct specification, we have $\lambda_0 = 0$ and $\delta = \delta_0$, and hence $t_i(\theta_0; \lambda_0, \delta_0) = (\delta_0 - \delta_0) + \lambda_0' g_i(\theta_0) = 0$, so that $t = 0$ is the appropriate expansion point. By Theorem 3.3, $t_i(\theta_0) = O_p(n^{-1/2})$, and hence $t_i(\theta_0) \xrightarrow{p} 0$. Therefore a Taylor expansion of $(1-\gamma t)^{1/\gamma}$ around $t = 0$ is justified. Moreover, $t_i^3 = O_p(n^{-3/2}) = o_p(n^{-1/2})$, so the cubic remainder term is negligible at the $\sqrt{n}$-scale relevant for first-order asymptotics. Consequently,

$$w_i = (1-\gamma t_i)^{1/\gamma} = 1 - t_i + \frac{1-\gamma}{2}\, t_i^2 + O(t_i^3), \tag{A.6}$$

where the expansion holds uniformly on events of probability approaching one under the stated remainder control.

Apply the probability constraint $\Psi_{n,1}(\hat\theta, \hat\lambda, \hat\delta) = 0$: $0 = \frac{1}{n}\sum_{i=1}^n w_i - 1 = \frac{1}{n}\sum_{i=1}^n \left(1 - t_i + \frac{1-\gamma}{2} t_i^2\right) - 1 + O\big(\frac{1}{n}\sum_{i=1}^n |t_i|^3\big)$. Cancelling the constants yields[6]

$$\hat\delta - \delta_0 = -\hat\lambda'\bar g(\hat\theta) + \frac{1-\gamma}{2} \cdot \frac{1}{n}\sum_{i=1}^n t_i(\hat\theta)^2 + o_p(n^{-1}). \tag{A.7}$$

Since $\hat\delta - \delta_0 = O_p(n^{-1})$ and $\hat\lambda' g_i(\hat\theta) = O_p(n^{-1/2})$, we have $\frac{1}{n}\sum_{i=1}^n t_i(\hat\theta)^2 = \hat\lambda'\hat\Omega(\hat\theta)\hat\lambda + o_p(n^{-1})$. The first term in A.7, recall A.4, where $\tilde\rho_n = o_p(1)\big(O_p(n^{-1/2}) + O_p(n^{-1})\big) =$

[6] In the present setting, $t_i(\theta_0) = (\hat\delta - \delta_0) + \hat\lambda' g_i(\theta_0)$, where $\hat\lambda = O_p(n^{-1/2})$ and $\hat\delta - \delta_0 = O_p(n^{-1})$ by Theorem 3.3. Hence $|t_i| \le |\hat\delta - \delta_0| + |\hat\lambda' g_i(\theta_0)|$. Since $g_i(\theta_0) = O_p(1)$ and $E\|g(Z,\theta_0)\|^3 < \infty$ under the stated moment conditions, we obtain

$$|t_i|^3 \lesssim |\hat\delta - \delta_0|^3 + |\hat\lambda' g_i(\theta_0)|^3 = O_p(n^{-3}) + O_p(n^{-3/2})\, \|g_i(\theta_0)\|^3.$$

Averaging over i yields

$$\frac{1}{n}\sum_{i=1}^n |t_i|^3 = O_p(n^{-3}) + O_p(n^{-3/2}) \left(\frac{1}{n}\sum_{i=1}^n \|g_i(\theta_0)\|^3\right).$$

By the law of large numbers, $\frac{1}{n}\sum_{i=1}^n \|g_i(\theta_0)\|^3 = O_p(1)$, and therefore $\frac{1}{n}\sum_{i=1}^n |t_i|^3 = O_p(n^{-3/2})$. In particular, this remainder term is $o_p(n^{-1/2})$ and hence negligible at the $\sqrt{n}$-scale relevant for the first-order asymptotic expansion.

$o_p(n^{-1/2})$, and $\tilde r_n = O_p\big(O_p(n^{-1}) + O_p(n^{-2})\big) = O_p(n^{-1}) = o_p(n^{-1/2})$. Hence both $\tilde\rho_n$ and $\tilde r_n$ are negligible at the $\sqrt{n}$-scale in the moment constraint expansion and $\hat\Omega(\hat\theta)\hat\lambda = \bar g(\hat\theta) + o_p(n^{-1/2})$. Thus, $\hat\lambda' \bar g(\theta) = \hat\lambda'\hat\Omega(\hat\theta)\hat\lambda + o_p(n^{-1})$. Therefore, $\hat\delta - \delta_0 = \left(\frac{1-\gamma}{2} - 1\right)\hat\lambda'\hat\Omega(\hat\theta)\hat\lambda + o_p(n^{-1}) = -\frac{\gamma+1}{2}\hat\lambda'\hat\Omega(\theta_0)\hat\lambda + o_p(n^{-1})$. Using $\hat\lambda = \hat\Omega(\theta_0)^{-1}\bar g(\theta_0) + o_p(n^{-1/2})$ and $\hat\Omega(\theta_0) \xrightarrow{p} \Omega_0$ yields

$$
\begin{aligned}
n(\hat\delta - \delta_0) &= -\frac{\gamma+1}{2}\left(\sqrt{n}\,\bar g(\theta_0)\right)'\Omega_0^{-1}\left(\sqrt{n}\,\bar g(\theta_0)\right) + o_p(1),\\
&= -\frac{\gamma+1}{2}\left(\Omega_0^{-1/2}\sqrt{n}\bar g(\theta_0)\right)'\left(\Omega_0^{-1/2}\sqrt{n}\bar g(\theta_0)\right) + o_p(1) \xrightarrow{d} -\frac{\gamma+1}{2}\chi^2_q,
\end{aligned}
$$

because $\sqrt{n}\,\bar g(\theta_0) \xrightarrow{d} N(0,\Omega_0)$. □

A.8 Proof for Theorem 3.5

By the envelope theorem, the first-order condition for the profiled problem can be written as the θ-derivative of the Lagrangian evaluated at the inner optimizer:

$$
0 = \frac{\partial}{\partial\theta}\left[\lambda'\sum_{i=1}^n \pi_i g_i(\theta)\right]_{(\theta,\lambda,\delta)=(\hat\theta,\hat\lambda(\hat\theta),\hat\delta(\hat\theta))} = \sum_{i=1}^n \hat\pi_i(\hat\theta)\,G_i(\hat\theta)'\hat\lambda,
$$

where $G_i(\theta) := \partial g_i(\theta)/\partial\theta'$. Divide by n: $0 = \frac{1}{n}\sum_{i=1}^n \hat\pi_i(\hat\theta)\,G_i(\hat\theta)'\hat\lambda$. Since $\hat\pi_i(\hat\theta) = \frac{1}{n}w_i(\hat\theta,\hat\lambda,\hat\delta)$, we may write $0 = \frac{1}{n}\sum_{i=1}^n \frac{1}{n}w_i(\hat\theta,\hat\lambda,\hat\delta)\,G_i(\hat\theta)'\hat\lambda = \left(\frac{1}{n}\sum_{i=1}^n w_i(\hat\theta,\hat\lambda,\hat\delta)\,G_i(\hat\theta)\right)'\hat\lambda\cdot\frac{1}{n}$. Multiplying both sides by n (which does not change the equality) yields $0 = \left(\frac{1}{n}\sum_{i=1}^n w_i(\hat\theta,\hat\lambda,\hat\delta)\,G_i(\hat\theta)\right)'\hat\lambda$. Decompose $w_i = 1 + (w_i - 1)$ to obtain $0 = \left(\frac{1}{n}\sum_{i=1}^n G_i(\hat\theta)\right)'\hat\lambda + \left(\frac{1}{n}\sum_{i=1}^n (w_i - 1)G_i(\hat\theta)\right)'\hat\lambda$. By the weight expansion, $w_i - 1 = O_p(n^{-1/2})$ for each fixed i[7] and under Assumption 3.2, we have $(1/n)\sum_{i=1}^n \|G_i(\hat\theta)\| = O_p(1)$. Since $\hat\lambda = O_p(n^{-1/2})$, it follows that $\left\|\left(\frac{1}{n}\sum_{i=1}^n (w_i - 1)G_i(\hat\theta)\right)'\hat\lambda\right\| \le \|\hat\lambda\|\cdot\frac{1}{n}\sum_{i=1}^n |w_i - 1|\,\|G_i(\hat\theta)\| = O_p(n^{-1/2})\cdot O_p(n^{-1/2}) = O_p(n^{-1})$, and in particular this term is $o_p(n^{-1/2})$.

[7] From the first-order expansion of the weights, $w_i(\hat\theta,\hat\lambda,\hat\delta) = 1 - (\hat\delta - \delta_0) - \hat\lambda' g_i(\hat\theta) + r_{i,n}$, so that $w_i - 1 = -(\hat\delta - \delta_0) - \hat\lambda' g_i(\hat\theta) + r_{i,n}$. By the previously established rates $\hat\lambda = O_p(n^{-1/2})$, $\hat\delta - \delta_0 = O_p(n^{-1})$, $r_{i,n} = O_p(\|\hat\lambda\|^2 + |\hat\delta - \delta_0|^2) = O_p(n^{-1})$, and since for each fixed i we have $g_i(\hat\theta) = O_p(1)$, it follows that $w_i - 1 = O_p(n^{-1/2})$ for each fixed i. Hence the natural pointwise stochastic rate is $w_i = 1 + O_p(n^{-1/2})$ for each fixed i.

Therefore,

$$0 = \Big(\frac{1}{n}\sum_{i=1}^{n} G_i(\hat{\theta})\Big)'\hat{\lambda} + o_p(n^{-1/2}). \tag{A.8}$$

By the uniform law of large numbers (from the envelope condition) and consistency $\hat{\theta} \xrightarrow{p} \theta_0$, $\frac{1}{n}\sum_{i=1}^{n} G_i(\hat{\theta}) \xrightarrow{p} G_0 := E[G(Z,\theta_0)]$. Moreover, by the first-order expansion for the multiplier,

$$\hat{\lambda} = \hat{\Omega}(\theta_0)^{-1}\,\bar{g}(\theta_0) + o_p(n^{-1/2}), \tag{A.9}$$

where $\hat{\Omega}(\theta_0) := \frac{1}{n}\sum_{i=1}^{n} g_i(\theta_0)g_i(\theta_0)'$ and $\hat{\Omega}(\theta_0) \xrightarrow{p} \Omega_0 := E[g(Z,\theta_0)g(Z,\theta_0)']$. Substituting (A.9) into (A.8) and using $\frac{1}{n}\sum_i G_i(\hat{\theta}) = G_0 + o_p(1)$ yields

$$0 = \Big(G_0' + o_p(1)\Big)\Big(\Omega_0^{-1}\bar{g}(\theta_0) + o_p(n^{-1/2})\Big) + o_p(n^{-1/2}) = G_0'\Omega_0^{-1}\bar{g}(\theta_0) + o_p(n^{-1/2}). \tag{A.10}$$

To express the result in terms of $\bar{g}(\hat{\theta})$, use the mean-value expansion $\bar{g}(\hat{\theta}) = \bar{g}(\theta_0) + G_0(\hat{\theta}-\theta_0) + o_p(n^{-1/2})$, so that $\bar{g}(\theta_0) = \bar{g}(\hat{\theta}) - G_0(\hat{\theta}-\theta_0) + o_p(n^{-1/2})$. Plugging this into (A.10) gives $0 = G_0'\Omega_0^{-1}\Big(\bar{g}(\hat{\theta}) - G_0(\hat{\theta}-\theta_0)\Big) + o_p(n^{-1/2}) = G_0'\Omega_0^{-1}\bar{g}(\hat{\theta}) - G_0'\Omega_0^{-1}G_0(\hat{\theta} - \theta_0) + o_p(n^{-1/2})$, or equivalently,

$$G_0'\Omega_0^{-1}G_0(\hat{\theta}-\theta_0) = G_0'\Omega_0^{-1}\bar{g}(\hat{\theta}) + o_p(n^{-1/2}). \tag{A.11}$$

Since G_0 has full column rank and $\Omega_0 \succ 0$, the matrix $G_0'\Omega_0^{-1}G_0$ is nonsingular, and solving (A.11) yields the linear representation $\hat{\theta} - \theta_0 = \big(G_0'\Omega_0^{-1}G_0\big)^{-1}G_0'\Omega_0^{-1}\bar{g}(\hat{\theta}) + o_p(n^{-1/2})$. Using again $\bar{g}(\hat{\theta}) = \bar{g}(\theta_0) + o_p(n^{-1/2})$, we obtain the equivalent form

$$\sqrt{n}(\hat{\theta}-\theta_0) = \big(G_0'\Omega_0^{-1}G_0\big)^{-1}G_0'\Omega_0^{-1}\sqrt{n}\,\bar{g}(\theta_0) + o_p(1). \tag{A.12}$$

By the multivariate central limit theorem, $\sqrt{n}\,\bar{g}(\theta_0) \xrightarrow{d} N(0,\Omega_0)$. Premultiplying by the constant matrix $A := \big(G_0'\Omega_0^{-1}G_0\big)^{-1}G_0'\Omega_0^{-1}$, and applying Slutsky's theorem to (A.12) yields $\sqrt{n}(\hat{\theta}-\theta_0) \xrightarrow{d} N\Big(0,\ A\,\Omega_0\,A'\Big) = N\Big(0,\ \big(G_0'\Omega_0^{-1}G_0\big)^{-1}\Big)$. □

A.9 Proof for Lemma 3.3

Recall (A.6): $w_i = 1 - t_i + \frac{1-\gamma}{2}t_i^2 + O(t_i^3)$, where $t_i = O_p(n^{-1/2})$. Plugging into the weighted moment constraint gives $0 = \frac{1}{n}\sum_{i=1}^{n} w_i g_i(\hat{\theta}) = \bar{g}(\hat{\theta}) - \frac{1}{n}\sum_{i=1}^{n} t_i g_i(\hat{\theta}) + \frac{1-\gamma}{2}\frac{1}{n}\sum_{i=1}^{n} t_i^2 g_i(\hat{\theta}) + \frac{1}{n}\sum_{i=1}^{n} O(t_i^3)g_i(\hat{\theta})$. Since $t_i = (\hat{\delta}-\delta_0) + \hat{\lambda}'g_i(\hat{\theta})$ and $\hat{\delta}-\delta_0 = O_p(n^{-1})$, we have $\frac{1}{n}\sum_{i=1}^{n} t_i g_i(\hat{\theta}) = (\hat{\delta}-\delta_0)\bar{g}(\hat{\theta}) + \hat{\Omega}(\hat{\theta})\hat{\lambda}$. The term $(\hat{\delta}-\delta_0)\bar{g}(\hat{\theta}) = O_p(n^{-3/2})$

is negligible at order n^{-1}. For the remainder, $\frac{1}{n}\sum |t_i|^3 = O_p(n^{-3/2})$, so under the moment bounds, $\frac{1}{n}\sum O(t_i^3)g_i(\hat{\theta}) = o_p(n^{-1})$, yielding (6). $\square$

A.10 Proof for Theorem 3.6

Theorem 3.2 and (6) imply

$$\hat{\Omega}(\theta_0)\hat{\lambda} = \bar{g}(\theta_0) + \frac{1-\gamma}{2} \cdot \frac{1}{n}\sum_{i=1}^{n} t_i(\theta_0, \hat{\lambda}, \hat{\delta})^2\, g_i(\theta_0) + o_p(n^{-1}). \tag{A.13}$$

Moreover, $t_i(\theta_0, \hat{\lambda}, \hat{\delta}) = \hat{\lambda}' g_i(\theta_0) + O_p(n^{-1})$, so $\frac{1}{n}\sum_{i=1}^{n} t_i^2 g_i(\theta_0) = \frac{1}{n}\sum_{i=1}^{n}(\hat{\lambda}' g_i(\theta_0))^2 g_i(\theta_0) + o_p(n^{-1})$. Substituting the first-order representation $\hat{\lambda} = \hat{\Omega}(\theta_0)^{-1}\bar{g}(\theta_0) + o_p(n^{-1/2})$ into the quadratic term yields, uniformly in i on events of probability approaching one, $(\hat{\lambda}' g_i(\theta_0))^2 = \big((\hat{\Omega}(\theta_0)^{-1}\bar{g}(\theta_0))' g_i(\theta_0)\big)^2 + o_p(n^{-1})$, and therefore

$$\frac{1}{n}\sum_{i=1}^{n} t_i^2 g_i(\theta_0) = \frac{1}{n}\sum_{i=1}^{n}\big((\hat{\Omega}(\theta_0)^{-1}\bar{g}(\theta_0))' g_i(\theta_0)\big)^2 g_i(\theta_0) + o_p(n^{-1}). \tag{A.14}$$

Multiplying (A.13) by $\hat{\Omega}(\theta_0)^{-1}$ and collecting all terms of order n^{-1} gives

$$\hat{\lambda} = \hat{\Omega}(\theta_0)^{-1}\bar{g}(\theta_0) + \frac{1}{n}B_{\lambda,n}(\gamma) + o_p(n^{-1}), \tag{A.15}$$

where

$$B_{\lambda,n}(\gamma) = \frac{1-\gamma}{2}\,\hat{\Omega}(\theta_0)^{-1}\left[\frac{1}{n}\sum_{i=1}^{n}\big((\hat{\Omega}(\theta_0)^{-1}\sqrt{n}\,\bar{g}(\theta_0))' g_i(\theta_0)\big)^2 g_i(\theta_0)\right]. \tag{A.16}$$

To verify the stochastic order of the bracketed term in (A.16), write $v_n := \hat{\Omega}(\theta_0)^{-1}\sqrt{n}\,\bar{g}(\theta_0)$, so that the bracket equals $\frac{1}{n}\sum_{i=1}^{n}\big(v_n' g_i(\theta_0)\big)^2 g_i(\theta_0)$. Using the inequality $|v_n' g_i(\theta_0)| \leq \|v_n\|\,\|g_i(\theta_0)\|$, we obtain $\left\|\frac{1}{n}\sum_{i=1}^{n}\big(v_n' g_i(\theta_0)\big)^2 g_i(\theta_0)\right\| \leq \|v_n\|^2 \cdot \frac{1}{n}\sum_{i=1}^{n}\|g_i(\theta_0)\|^3$. Since $\hat{\Omega}(\theta_0)^{-1} = O_p(1)$ as $\hat{\Omega} \xrightarrow{p} \Omega_0 \succ 0$ and $\sqrt{n}\,\bar{g}(\theta_0) = O_p(1)$ by the central limit theorem, it follows that $\|v_n\| = O_p(1)$ and hence $\|v_n\|^2 = O_p(1)$. Moreover, under Assumption 3.2, the law of large numbers yields $\frac{1}{n}\sum_{i=1}^{n}\|g_i(\theta_0)\|^3 = O_p(1)$. Combining these bounds gives $\frac{1}{n}\sum_{i=1}^{n}\big(v_n' g_i(\theta_0)\big)^2 g_i(\theta_0) = O_p(1)$. Hence the bracketed term in (A.16) is $O_p(1)$. Thus, $B_{\lambda,n}(\gamma) = O_p(1)$. $\square$

A.11 Proof for Theorem 3.7

Step 1: First-order condition for θ.

Let $\hat{\pi}_i(\theta)$ denote the implied probability weights at $(\theta, \hat{\lambda}(\theta), \hat{\delta}(\theta))$. The envelope first-order condition (FOC) for θ is

$$
\begin{aligned}
0 \overset{\text{FOC}}{=} \left.\frac{\partial \mathcal{L}}{\partial \theta}\right|_{\pi=\hat{\pi}(\theta),\lambda=\hat{\lambda}(\theta),\delta=\hat{\delta}(\theta)} &= \hat{\lambda}' \sum_{i=1}^{n} \hat{\pi}_i(\hat{\theta}) \left.\frac{\partial g_i}{\partial \theta}\right|_{\theta=\hat{\theta}} \\
&= \sum_{i=1}^{n} \hat{\pi}_i(\hat{\theta}) \left.\frac{\partial g_i(\theta)}{\partial \theta}\right|_{\theta=\hat{\theta}}' \hat{\lambda} \\
&= \sum_{i=1}^{n} \hat{\pi}_i(\hat{\theta}) G_i(\hat{\theta})'\hat{\lambda}.
\end{aligned}
$$

Since $\hat{\pi}_i(\hat{\theta}) = \frac{1}{n} w_i(\hat{\theta}, \hat{\lambda}, \hat{\delta})$, this condition can be written $0 = \frac{1}{n}\sum_{i=1}^{n} w_i(\hat{\theta}, \hat{\lambda}, \hat{\delta})\, G_i(\hat{\theta})'\hat{\lambda}$. Define $\bar{G}(\theta) = \frac{1}{n}\sum_{i=1}^{n} G_i(\theta)$. Then the FOC becomes $0 = \bar{G}(\hat{\theta})'\hat{\lambda} + R_{w,G}$, where $R_{w,G} = \frac{1}{n}\sum_{i=1}^{n} \big(w_i(\hat{\theta}, \hat{\lambda}, \hat{\delta}) - 1\big) G_i(\hat{\theta})'\hat{\lambda}$.

Step 2: Order of the weight remainder.

From the multiplier expansion we have $\hat{\lambda} = O_p(n^{-1/2})$. Let $\psi_\gamma(u) := \left(\frac{1}{\gamma+1} - \gamma u\right)^{1/\gamma}$, with $u = \delta + \lambda' g_i(\theta)$. Under the smoothness conditions for the CRPD weights, $\frac{1}{n}\sum_{i=1}^{n} |w_i - 1| = O_p(n^{-1/2})$. To see why, because the CRPD weights are smooth in (λ, δ), a first-order Taylor expansion around the population solution $(\lambda, \delta) = (0, \delta_0)$ yields $w_i(\theta_0, \lambda, \delta) = w_i(\theta_0, 0, \delta_0) + \psi_\gamma'(u_0)\, \lambda' g_i(\theta_0) + O(\|\lambda\|^2)$.[8]

[8] Fix $\theta = \theta_0$ and write the CRPD weight in terms of the scalar index

$$u_i(\lambda, \delta) := \delta + \lambda' g_i(\theta_0), \qquad w_i(\theta_0, \lambda, \delta) = \psi_\gamma\big(u_i(\lambda, \delta)\big),$$

where $\psi_\gamma(u) := \left(\frac{1}{\gamma+1} - \gamma u\right)^{1/\gamma}$. Let $(\lambda_0, \delta_0) = (0, \delta_0)$ denote the population solution and set $u_0 := u_i(0, \delta_0) = \delta_0$.

Step 1 (Taylor expansion in the scalar index). Since $\psi_\gamma(\cdot)$ is differentiable at u_0, Taylor's theorem implies $\psi_\gamma(u) = \psi_\gamma(u_0) + \psi_\gamma'(u_0)(u - u_0) + O\big((u - u_0)^2\big),\ u \to u_0$.

Applying this with $u = u_i(\lambda, \delta)$ gives

$$w_i(\theta_0, \lambda, \delta) = w_i(\theta_0, 0, \delta_0) + \psi_\gamma'(u_0)\Big[(\delta - \delta_0) + \lambda' g_i(\theta_0)\Big] + O\left(\Big|(\delta - \delta_0) + \lambda' g_i(\theta_0)\Big|^2\right).$$

Step 2 (Control the remainder). Using $(a+b)^2 \le 2a^2 + 2b^2$, $\Big((\delta - \delta_0) + \lambda' g_i(\theta_0)\Big)^2 \le 2(\delta -$

Under correct specification, $\lambda_0 = 0$ and $\delta_0 = u_0 = -1/(\gamma+1)$, so $w_i(\theta_0, 0, \delta_0) = \psi_\gamma(u_0) = 1$ and $\psi'_\gamma(u_0) = -1$. Hence $w_i(\theta_0, \lambda, \delta) = 1 - \lambda' g_i(\theta_0) + O(\|\lambda\|^2)$. Hence $|w_i - 1| \leq C\,|\lambda' g_i(\theta_0)| + O(\|\lambda\|^2)$ for some constant $C > 0$.

Since $\lambda = O_p(n^{-1/2})$, $|w_i - 1| = O_p(n^{-1/2})$ for each i. Averaging across i and using $E\|g(Z,\theta_0)\| < \infty$ gives $\frac{1}{n}\sum_{i=1}^n |w_i - 1| = O_p(n^{-1/2})$. Moreover $\frac{1}{n}\sum_{i=1}^n \|G_i(\hat\theta)\| = O_p(1)$. Therefore $\|R_{w,G}\| \leq \left(\frac{1}{n}\sum_{i=1}^n |w_i - 1|\,\|G_i(\hat\theta)\|\right)\|\hat\lambda\| = O_p(n^{-1})$. Thus $0 = \bar G(\hat\theta)'\hat\lambda + O_p(n^{-1})$.

Step 3: Taylor expansion of $\bar G(\hat\theta)$.

Expand $\bar G(\hat\theta)$ around θ_0: $\bar G(\hat\theta) = \bar G(\theta_0) + \bar H(\theta_0)(\hat\theta - \theta_0) + R_G$, where $\bar H(\theta_0) = \frac{1}{n}\sum_{i=1}^n H_i(\theta_0)$, $\|R_G\| = O_p(\|\hat\theta - \theta_0\|^2)$. Since $\hat\theta - \theta_0 = O_p(n^{-1/2})$, the remainder satisfies $R_G = O_p(n^{-1})$. Substituting into the first-order condition gives $0 = \Big(\bar G(\theta_0) + \bar H(\theta_0)(\hat\theta - \theta_0)\Big)'\hat\lambda + O_p(n^{-1})$.

Step 4: Expansion of $\hat\lambda$.

From Theorem 3.6, $\hat\lambda = \hat\Omega(\theta_0)^{-1}\bar g(\theta_0) + \frac{1}{n}B_{\lambda,n}(\gamma) + o_p(n^{-1})$, with $B_{\lambda,n}(\gamma) = O_p(1)$. Substituting this expansion gives

$$0 = \bar G(\theta_0)'\hat\Omega(\theta_0)^{-1}\bar g(\theta_0) + \bar H(\theta_0)'(\hat\theta - \theta_0)\hat\Omega(\theta_0)^{-1}\bar g(\theta_0) + \frac{1}{n}\bar G(\theta_0)'B_{\lambda,n}(\gamma) + o_p(n^{-1}), \tag{A.17}$$

with $\frac{1}{n}\bar H(\theta_0)'(\hat\theta - \theta_0)B_{\lambda,n} = O_p(n^{-3/2}) = o_p(n^{-1})$.

Step 5: Replace sample matrices with population limits.

Since $\hat\theta - \theta_0 = O_p(n^{-1/2})$, $\bar g(\theta_0) = O_p(n^{-1/2})$, $\hat\Omega(\theta_0)^{-1} = O_p(1)$, and $\bar H(\theta_0) = O_p(1)$, we have $\bar H(\theta_0)'(\hat\theta - \theta_0)\hat\Omega(\theta_0)^{-1}\bar g(\theta_0) = O_p(n^{-1})$. Hence we can absorb this term into the order-n^{-1} remainder by defining a new random scalar $\widetilde\Delta_{H,n}$ such that

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$\delta_0)^2 + 2\big(\lambda' g_i(\theta_0)\big)^2$. Under the maintained rates $\lambda = O_p(n^{-1/2})$ and $\delta - \delta_0 = O_p(n^{-1})$, and assuming $E\|g(Z,\theta_0)\|^2 < \infty$, we have $(\delta - \delta_0)^2 = O_p(n^{-2}) = o_p(\|\lambda\|^2)$, $\big(\lambda' g_i(\theta_0)\big)^2 = O_p(\|\lambda\|^2)$. Hence $O\left(\left|(\delta - \delta_0) + \lambda' g_i(\theta_0)\right|^2\right) = O_p(\|\lambda\|^2)$.

Step 3 (Absorb the δ term). Moreover, $\psi'_\gamma(u_0)(\delta - \delta_0) = O_p(n^{-1}) = o_p(\|\lambda\|)$, so this term is of smaller order than the $\lambda' g_i(\theta_0$ term and can be absorbed into the $O_p(\|\lambda\|^2)$ remainder at the scale of interest.

Step 4 (Simplified expansion). Combining the above yields $w_i(\theta_0, \lambda, \delta) = w_i(\theta_0, 0, \delta_0) + \psi'_\gamma(u_0)\,\lambda' g_i(\theta_0) + O_p(\|\lambda\|^2)$, which is the desired form.

$\bar{H}(\theta_0)'(\hat{\theta}-\theta_0)\hat{\Omega}(\theta_0)^{-1}\bar{g}(\theta_0) = \frac{1}{n}\widetilde{\Delta}_{H,n} + o_p(n^{-1})$, $\widetilde{\Delta}_{H,n} = O_p(1)$. Then (A.17) becomes

$$0 = \bar{G}(\theta_0)'\hat{\Omega}(\theta_0)^{-1}\bar{g}(\theta_0) + \frac{1}{n}\bar{G}(\theta_0)'B_{\lambda,n}(\gamma) + \frac{1}{n}\widetilde{\Delta}_{H,n} + o_p(n^{-1}). \tag{A.18}$$

Moreover, a first-order Taylor expansion of the sample moment yields

$$\bar{g}(\hat{\theta}) = \bar{g}(\theta_0)+\bar{G}(\theta_0)(\hat{\theta}-\theta_0)+O_p(\|\hat{\theta}-\theta_0\|^2) = \bar{g}(\theta_0)+\bar{G}(\theta_0)(\hat{\theta}-\theta_0)+O_p(n^{-1}). \tag{A.19}$$

Premultiplying by $\bar{G}(\theta_0)'\hat{\Omega}(\theta_0)^{-1}$ gives

$$\bar{G}(\theta_0)'\hat{\Omega}(\theta_0)^{-1}\bar{g}(\hat{\theta}) = \bar{G}(\theta_0)'\hat{\Omega}(\theta_0)^{-1}\bar{g}(\theta_0) + \bar{G}(\theta_0)'\hat{\Omega}(\theta_0)^{-1}\bar{G}(\theta_0)(\hat{\theta}-\theta_0) + O_p(n^{-1}). \tag{A.20}$$

Rearranging (A.20) yields the identity

$$\bar{G}(\theta_0)'\hat{\Omega}(\theta_0)^{-1}\bar{g}(\theta_0) = \bar{G}(\theta_0)'\hat{\Omega}(\theta_0)^{-1}\bar{g}(\hat{\theta}) - \bar{G}(\theta_0)'\hat{\Omega}(\theta_0)^{-1}\bar{G}(\theta_0)(\hat{\theta}-\theta_0) + O_p(n^{-1}). \tag{A.21}$$

Substituting (A.21) into (A.18) gives

$$0 = \bar{G}(\theta_0)'\hat{\Omega}(\theta_0)^{-1}\bar{g}(\hat{\theta}) - \bar{G}(\theta_0)'\hat{\Omega}(\theta_0)^{-1}\bar{G}(\theta_0)(\hat{\theta}-\theta_0) + \frac{1}{n}\bar{G}(\theta_0)'B_{\lambda,n}(\gamma) + \frac{1}{n}\widetilde{\Delta}_{H,n} + o_p(n^{-1}). \tag{A.22}$$

(i) First term. Since $\bar{G}(\theta_0) = G_0 + O_p(n^{-1/2})$ and $\hat{\Omega}(\theta_0)^{-1} = \Omega_0^{-1} + O_p(n^{-1/2})$, we have $\bar{G}(\theta_0)'\hat{\Omega}(\theta_0)^{-1} = (G_0 + O_p(n^{-1/2}))'(\Omega_0^{-1} + O_p(n^{-1/2})) = G_0'\Omega_0^{-1} + O_p(n^{-1/2})$. Moreover, $\bar{g}(\hat{\theta}) = O_p(n^{-1/2})$ under the maintained assumptions. Thus,

$$\bar{G}(\theta_0)'\hat{\Omega}(\theta_0)^{-1}\bar{g}(\hat{\theta}) = G_0'\Omega_0^{-1}\bar{g}(\hat{\theta}) + O_p(n^{-1}). \tag{A.23}$$

(ii) Second term. Using $\bar{G}(\theta_0) = G_0 + O_p(n^{-1/2})$ and $\hat{\Omega}(\theta_0)^{-1} = \Omega_0^{-1} + O_p(n^{-1/2})$,

$$\begin{aligned}\bar{G}(\theta_0)'\hat{\Omega}(\theta_0)^{-1}\bar{G}(\theta_0) &= (G_0 + O_p(n^{-1/2}))'(\Omega_0^{-1} + O_p(n^{-1/2}))(G_0 + O_p(n^{-1/2})) \\ &= G_0'\Omega_0^{-1}G_0 + O_p(n^{-1/2}),\end{aligned}$$

and therefore, since $\hat{\theta} - \theta_0 = O_p(n^{-1/2})$,

$$\bar{G}(\theta_0)'\hat{\Omega}(\theta_0)^{-1}\bar{G}(\theta_0)(\hat{\theta} - \theta_0) = G_0'\Omega_0^{-1}G_0(\hat{\theta} - \theta_0) + O_p(n^{-1}). \tag{A.24}$$

(iii) Third term. Since $\bar{G}(\theta_0) = G_0 + O_p(n^{-1/2})$ and $B_{\lambda,n}(\gamma) = O_p(1)$,

$$\begin{aligned} \frac{1}{n}\bar{G}(\theta_0)'B_{\lambda,n}(\gamma) &= \frac{1}{n}G_0'B_{\lambda,n}(\gamma) + \frac{1}{n}O_p(n^{-1/2})O_p(1) \\ &= \frac{1}{n}G_0'B_{\lambda,n}(\gamma) + o_p(n^{-1}). \end{aligned} \tag{A.25}$$

(iv) Fourth term. By construction, $\widetilde{\Delta}_{H,n} = O_p(1)$, hence

$$\frac{1}{n}\widetilde{\Delta}_{H,n} = O_p(n^{-1}). \tag{A.26}$$

Substitution into (A.22)**.** Substituting (A.23)–(A.26) into (A.22) yields $0 = G_0'\Omega_0^{-1}\bar{g}(\hat{\theta}) - G_0'\Omega_0^{-1}G_0(\hat{\theta} - \theta_0) + \frac{1}{n}G_0'B_{\lambda,n}(\gamma) + \frac{1}{n}\widetilde{\Delta}_{H,n} + O_p(n^{-1}) + o_p(n^{-1})$. Absorbing the $O_p(n^{-1})$ terms into a single $\frac{1}{n}\Delta_n(\gamma)$ term gives $0 = G_0'\Omega_0^{-1}\bar{g}(\hat{\theta}) - G_0'\Omega_0^{-1}G_0(\hat{\theta} - \theta_0) + \frac{1}{n}\Delta_n(\gamma) + o_p(n^{-1})$, where $\Delta_n(\gamma) = O_p(1)$ collects all order-n^{-1} terms, including $G_0'B_{\lambda,n}(\gamma)$, $\widetilde{\Delta}_{H,n}$, and additional plug-in terms arising from replacing the sample matrices $\bar{G}(\theta_0)$ and $\hat{\Omega}(\theta_0)^{-1}$ by their population limits G_0 and Ω_0^{-1}.[9]

Since $G_0'\Omega_0^{-1}G_0$ is nonsingular, we can rearrange the preceding equation to obtain $G_0'\Omega_0^{-1}G_0(\hat{\theta} - \theta_0) = G_0'\Omega_0^{-1}\bar{g}(\hat{\theta}) + \frac{1}{n}\Delta_n(\gamma) + o_p(n^{-1})$. Premultiplying both sides by $\left(G_0'\Omega_0^{-1}G_0\right)^{-1}$ yields $\hat{\theta} - \theta_0 = \left(G_0'\Omega_0^{-1}G_0\right)^{-1}G_0'\Omega_0^{-1}\bar{g}(\hat{\theta}) + \frac{1}{n}\left(G_0'\Omega_0^{-1}G_0\right)^{-1}\Delta_n(\gamma) + o_p(n^{-1})$. The dependence of $\Delta_n(\gamma)$ on γ arises through the second-order expansion of the Lagrange multiplier $B_{\lambda,n}(\gamma)$, which depends on γ, as well as through the curvature term $\widetilde{\Delta}_{H,n}$ that interacts with the multiplier expansion. Consequently, $\Delta_n(\gamma)$ captures how γ affects the n^{-1} (second-order) behavior of the estimator. □

[9] These plug-in terms are generated by the deviations $\bar{G}(\theta_0) - G_0$ and $\hat{\Omega}(\theta_0)^{-1} - \Omega_0^{-1}$, both of which are $O_p(n^{-1/2})$, and therefore contribute $O_p(n^{-1})$ when multiplied by $O_p(n^{-1/2})$ quantities such as $\bar{g}(\theta_0)$ or $(\hat{\theta} - \theta_0)$.